Chapter 21

Time Series Econometrics: Some Basic Concepts

We noted in Chapter 1 that one of the important types of data used in empirical analysis is **time series** data. In this and the following chapter we take a closer look at such data not only because of the frequency with which they are used in practice but also because they pose several challenges to econometricians and practitioners.

First, empirical work based on time series data assumes that the underlying time series is **stationary.** Although we have discussed the concept of stationarity intuitively in Chapter 1, we discuss it more fully in this chapter. More specifically, we will try to find out what stationarity means and why one should worry about it.

Second, in Chapter 12, on autocorrelation, we discussed several causes of autocorrelation. Sometimes autocorrelation results because the underlying time series is nonstationary.

Third, in regressing a time series variable on another time series variable(s), one often obtains a very high R^2 (in excess of 0.9) even though there is no meaningful relationship between the two variables. Sometimes we expect no relationship between two variables, yet a regression of one on the other variable often shows a significant relationship. This situation exemplifies the problem of **spurious,** or **nonsense, regression,** whose nature will be explored shortly. It is therefore very important to find out if the relationship between economic variables is spurious or nonsensical. We will see in this chapter how spurious regressions can arise if time series are not stationary.

Fourth, some financial time series, such as stock prices, exhibit what is known as the **random walk phenomenon.** This means the best prediction of the price of a stock, say IBM, tomorrow is equal to its price today plus a purely random shock (or error term). If this were in fact the case, forecasting asset prices would be a futile exercise.

Fifth, regression models involving time series data are often used for forecasting. In view of the preceding discussion, we would like to know if such forecasting is valid if the underlying time series are not stationary.

Finally, causality tests (recall the Granger and Sims causality tests discussed in Chapter 17) assume that the time series involved in analysis are stationary. Therefore, tests of stationarity should precede tests of causality.

At the outset a disclaimer is in order. The topic of time series analysis is so vast and evolving and some of the mathematics underlying the various techniques of time series analysis is so involved that the best we hope to achieve in an introductory text like this is to

give the reader a glimpse of some of the fundamental concepts of time series analysis. For those who want to pursue this topic further, we provide references.[1]

21.1 A Look at Selected U.S. Economic Time Series

To set the ball rolling, and to give the reader a feel for the somewhat esoteric concepts of time series analysis to be developed in this chapter, it might be useful to consider several U.S. economic time series of general interest. The time series we consider are:

DPI = real disposable personal income (billions of dollars)
GDP = gross domestic product (billions of dollars)
PCE = real personal consumption expenditure (billions of dollars)
CP = corporate profits (billions of dollars)
Dividend = dividends, (billions of dollars)

The time period covered is from 1947–I to 2007–IV, for a total of 244 quarters, and all data are seasonally adjusted at the annual rate. All the data are collected from FRED, the economic website of the Federal Reserve Bank of St. Louis. GDP, DPI, and PCE are in constant dollars, here 2000 dollars. CP and Dividend are in nominal dollars.

To save space, the raw data are posted on the book's website. But to get some idea of these data, we have plotted them in the following two figures. Figure 21.1 is a plot of the data of logarithms of GDP, DPI, and PCE and Figure 21.2 presents the logs of the other two time series (CP and Dividend). It is common practice to plot the log of a time series to get a glimpse of the growth rate of such a series. A visual plot of the data is usually the first step in the analysis of time series. In these figures the letter L denotes the natural logarithm.

The first impression we get from these two figures is that all these time series seem to be "trending" upward, albeit with fluctuations. Suppose we want to speculate on the shape of these curves beyond the sample period, say for all the quarters of 2008.[2] We can do that if we know the statistical, or stochastic, mechanism, or the **data generating process (DGP)** that generated these curves. But what is that mechanism? To answer this and related questions, we need to study some "new" vocabulary that has been developed by time series analysts, to which we now turn.

[1]At the introductory level, these references may be helpful: Gary Koop, *Analysis of Economic Data,* John Wiley & Sons, New York, 2000; Jeff B. Cromwell, Walter C. Labys, and Michel Terraza, *Univariate Tests for Time Series Models,* Sage Publications, California, Ansbury Park, 1994; Jeff B. Cromwell, Michael H. Hannan, Walter C. Labys, and Michel Terraza, *Multivariate Tests for Time Series Models,* Sage Publications, California, Ansbury Park, 1994; and H. R. Seddighi, K. A. Lawler, and A. V. Katos, *Econometrics: A Practical Approach,* Routledge, New York, 2000. At the intermediate level, see Walter Enders, *Applied Econometric Time Series,* John Wiley & Sons, New York, 1995; Kerry Patterson, *An Introduction to Applied Econometrics: A Time Series Approach,* St. Martin's Press, New York, 2000; T. C. Mills, *The Econometric Modelling of Financial Time Series,* 2d ed., Cambridge University Press, New York, 1999; Marno Verbeek, *A Guide to Modern Econometrics,* John Wiley & Sons, New York, 2000; and Wojciech W. Charemza and Derek F. Deadman, *New Directions in Econometric Practice: General to Specific Modelling and Vector Autoregression,* 2d ed., Edward Elgar Publisher, New York, 1997. At the advanced level, see J. D. Hamilton, *Time Series Analysis,* Princeton University Press, Princeton, NJ, 1994, and G. S. Maddala and In-Moo Kim, *Unit Roots, Cointegration, and Structural Change,* Cambridge University Press, 1998. At the applied level, see B. Bhaskara Rao, ed., *Cointegration for the Applied Economist,* St. Martin's Press, New York, 1994, and Chandan Mukherjee, Howard White, and Marc Wuyts, *Econometrics and Data Analysis for Developing Countries,* Routledge, New York, 1998.

[2]Of course, we have the actual data for this period now and could compare it with the data that is "predicted" on the basis of the earlier period.

FIGURE 21.1
Logarithms of real GDP, DPI, and PCE, United States, 1947–2007 (quarterly, $ billions).

Note: In the figure the letter L denotes natural logarithm.

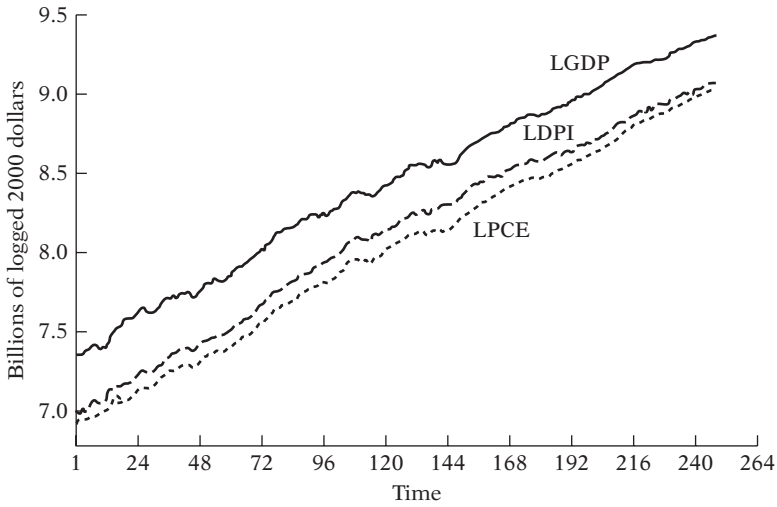
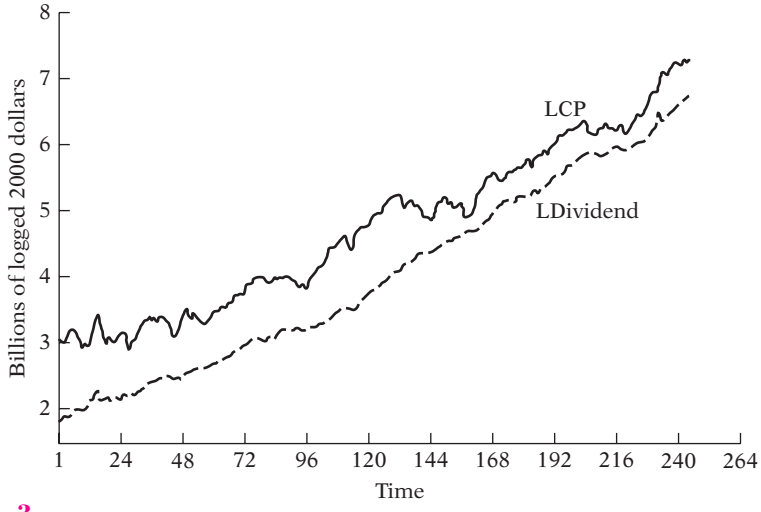


FIGURE 21.2
Logarithms of corporate profits (CP) and dividends, United States, 1947–2007 (quarterly, $ billions).

Note: L denotes logarithm.



21.2 Key Concepts[3]

What is this vocabulary? It consists of concepts such as these:

1. Stochastic processes
2. Stationarity processes
3. Purely random processes
4. Nonstationary processes
5. Integrated variables
6. Random walk models
7. Cointegration
8. Deterministic and stochastic trends
9. Unit root tests

In what follows we will discuss each of these concepts. Our discussion will often be heuristic. Wherever possible and helpful, we will provide appropriate examples.

[3]The following discussion is based on Maddala et al., op. cit., Charemza et al., op. cit., and Carol Alexander, *Market Models: A Guide to Financial Data Analysis*, John Wiley & Sons, New York, 2001.

21.3 Stochastic Processes

A random or stochastic process is a collection of random variables ordered in time.[4] If we let Y denote a random variable, and if it is continuous, we denote it as $Y(t)$, but if it is discrete, we denoted it as Y_t. An example of the former is an electrocardiogram, and an example of the latter is GDP, DPI, etc. Since most economic data are collected at discrete points in time, for our purpose we will use the notation Y_t rather than $Y(t)$. If we let Y represent GDP, for our data we have $Y_1, Y_2, Y_3, \ldots, Y_{242}, Y_{243}, Y_{244}$, where the subscript 1 denotes the first observation (i.e., GDP for the first quarter of 1947) and the subscript 244 denotes the last observation (i.e., GDP for the fourth quarter of 2007). *Keep in mind that each of these Y's is a random variable.*

In what sense can we regard GDP as a stochastic process? Consider for instance the real GDP of $3,759.997 billion for 1970–I. In theory, the GDP figure for the first quarter of 1970 could have been any number, depending on the economic and political climate then prevailing. The figure of 3,759.997 is a particular **realization** of all such possibilities.[5] Therefore, we can say that GDP is a stochastic process and the actual values we observed for the period 1947–I to 2007–IV are particular realizations of that process (i.e., sample). The distinction between the stochastic process and its realization is akin to the distinction between population and sample in cross-sectional data. Just as we use sample data to draw inferences about a population, in time series we use the realization to draw inferences about the underlying stochastic process.

Stationary Stochastic Processes

A type of stochastic process that has received a great deal of attention and scrutiny by time series analysts is the so-called **stationary stochastic process.** Broadly speaking, *a stochastic process is said to be stationary if its mean and variance are constant over time and the value of the covariance between the two time periods depends only on the distance or gap or lag between the two time periods and not the actual time at which the covariance is computed.* In the time series literature, such a stochastic process is known as a **weakly stationary,** or **covariance stationary,** or **second-order stationary,** or **wide sense, stochastic process.** For the purpose of this chapter, and in most practical situations, this type of stationarity often suffices.[6]

To explain weak stationarity, let Y_t be a stochastic time series with these properties:

$$\text{Mean:} \qquad E(Y_t) = \mu \tag{21.3.1}$$

$$\text{Variance:} \qquad \text{var}\,(Y_t) = E(Y_t - \mu)^2 = \sigma^2 \tag{21.3.2}$$

$$\text{Covariance:} \qquad \gamma_k = E[(Y_t - \mu)(Y_{t+k} - \mu)] \tag{21.3.3}$$

where γ_k, the covariance (or autocovariance) at lag k, is the covariance between the values of Y_t and Y_{t+k}, that is, between two Y values k periods apart. If $k = 0$, we obtain γ_0, which

[4]The term "stochastic" comes from the Greek word "stokhos," which means a target or bull's-eye. If you have ever thrown darts on a dart board with the aim of hitting the bull's-eye, how often did you hit the bull's-eye? Out of a hundred darts you may be lucky to hit the bull's-eye only a few times; at other times the darts will be spread randomly around the bull's-eye.

[5]You can think of the value of $3,759.997 billion as the mean value of all possible values of GDP for the first quarter of 1970.

[6]A time series is strictly stationary if *all* the moments of its probability distribution and not just the first two (i.e., mean and variance) are invariant over time. If, however, the stationary process is normal, the weakly stationary stochastic process is also strictly stationary, for the normal stochastic process is fully specified by its two moments, the mean and the variance.

is simply the variance of Y ($= \sigma^2$); if $k = 1$, γ_1 is the covariance between two adjacent values of Y, the type of covariance we encountered in Chapter 12 (recall the Markov first-order autoregressive scheme).

Suppose we shift the origin of Y from Y_t to Y_{t+m} (say, from the first quarter of 1947 to the first quarter of 1952 for our GDP data). Now if Y_t is to be stationary, the mean, variance, and autocovariances of Y_{t+m} must be the same as those of Y_t. *In short, if a time series is stationary, its mean, variance, and autocovariance (at various lags) remain the same no matter at what point we measure them; that is, they are time invariant.* Such a time series will tend to return to its mean (called **mean reversion**) and fluctuations around this mean (measured by its variance) will have a broadly constant amplitude.[7] To put it differently, a stationary process will not drift too far away from its mean value because of the finite variance. As we shall see shortly, this is not the case with nonstationary stochastic processes. It should be noted that for a stationary process the speed of mean reversion depends on the autocovariances; it is quick if the autocovariances are small and slow when they are large, as we will show shortly.

If a time series is not stationary in the sense just defined, it is called a **nonstationary time series** (keep in mind we are talking only about weak stationarity). In other words, a nonstationary time series will have a *time-varying mean or a time-varying variance or both.*

Why are stationary time series so important? Because if a time series is nonstationary, we can study its behavior only for the time period under consideration. Each set of time series data will therefore be for a particular episode. As a consequence, it is not possible to generalize it to other time periods. Therefore, for the purpose of forecasting, such (nonstationary) time series may be of little practical value.

How do we know that a particular time series is stationary? In particular, are the time series shown in Figures 21.1 and 21.2 stationary? We will take this important topic up in Sections 21.8 and 21.9, where we will consider several tests of stationarity. But if we depend on common sense, it would seem that the time series depicted in Figures 21.1 and 21.2 are nonstationary, at least in the mean values. But more on this later.

Before we move on, we mention a special type of stochastic process (or time series), namely, a **purely random,** or **white noise, process.** We call a stochastic process purely random if it has zero mean, constant variance σ^2, and is serially uncorrelated.[8] You may recall that the error term u_t, entering the classical normal linear regression model that we discussed in **Part 1** of this book, was assumed to be a white noise process, which we denoted as $u_t \sim \text{IIDN}(0, \sigma^2)$; that is, u_t is independently and identically distributed as a normal distribution with zero mean and constant variance. Such a process is called a **Gaussian white noise process.**

Nonstationary Stochastic Processes

Although our interest is in stationary time series, one often encounters nonstationary time series, the classic example being the **random walk model** (RWM).[9] It is often said that asset prices, such as stock prices or exchange rates, follow a random walk; that is, they are nonstationary. We distinguish two types of random walks: (1) random walk without drift (i.e., no constant or intercept term) and (2) random walk with drift (i.e., a constant term is present).

[7]This point has been made by Keith Cuthbertson, Stephen G. Hall, and Mark P. Taylor, *Applied Econometric Techniques,* The University of Michigan Press, 1995, p. 130.

[8]If it is also independent, such a process is called **strictly white noise.**

[9]The term random walk is often compared with a drunkard's walk. Leaving a bar, the drunkard moves a random distance u_t at time t, and, continuing to walk indefinitely, will eventually drift farther and farther away from the bar. The same is said about stock prices. Today's stock price is equal to yesterday's stock price plus a random shock.

Random Walk without Drift

Suppose u_t is a white noise error term with mean 0 and variance σ^2. Then the series Y_t is said to be a random walk if

$$Y_t = Y_{t-1} + u_t \tag{21.3.4}$$

In the random walk model, as Eq. (21.3.4) shows, the value of Y at time t is equal to its value at time $(t-1)$ plus a random shock; thus it is an AR(1) model in the language of Chapters 12 and 17. We can think of Eq. (21.3.4) as a regression of Y at time t on its value lagged one period. Believers in the **efficient capital market hypothesis** argue that stock prices are essentially random and therefore there is no scope for profitable speculation in the stock market: If one could predict tomorrow's price on the basis of today's price, we would all be millionaires.

Now from Eq. (21.3.4) we can write

$$Y_1 = Y_0 + u_1$$
$$Y_2 = Y_1 + u_2 = Y_0 + u_1 + u_2$$
$$Y_3 = Y_2 + u_3 = Y_0 + u_1 + u_2 + u_3$$

In general, if the process started at some time 0 with a value of Y_0, we have

$$Y_t = Y_0 + \sum u_t \tag{21.3.5}$$

Therefore,

$$E(Y_t) = E\left(Y_0 + \sum u_t\right) = Y_0 \quad \text{(why?)} \tag{21.3.6}$$

In like fashion, it can be shown that

$$\text{var}\,(Y_t) = t\sigma^2 \tag{21.3.7}$$

As the preceding expression shows, the mean of Y is equal to its initial, or starting, value, which is constant, but as t increases, its variance increases indefinitely, thus violating a condition of stationarity. In short, the RWM without drift is a nonstationary stochastic process. In practice Y_0 is often set at zero, in which case $E(Y_t) = 0$.

An interesting feature of the RWM is the *persistence of random shocks* (i.e., random errors), which is clear from Eq. (21.3.5): Y_t is the sum of initial Y_0 plus the sum of random shocks. As a result, the impact of a particular shock does not die away. For example, if $u_2 = 2$ rather than $u_2 = 0$, then all Y_t's from Y_2 onward will be 2 units higher and the effect of this shock never dies out. That is why random walk is said to have an *infinite memory*. As Kerry Patterson notes, random walk remembers the shock forever;[10] that is, it has infinite memory. The sum $\sum u_t$ is also known as a **stochastic trend,** about which more will be said shortly.

Interestingly, if you write Eq. (21.3.4) as

$$(Y_t - Y_{t-1}) = \Delta Y_t = u_t \tag{21.3.8}$$

where Δ is the first difference operator that we discussed in Chapter 12, it is easy to show that, while Y_t is nonstationary, its first difference is stationary. In other words, the first differences of a random walk time series are stationary. But we will have more to say about this later.

[10]Kerry Patterson, op cit., Chapter 6.

Random Walk with Drift

Let us modify Eq. (21.3.4) as follows:

$$Y_t = \delta + Y_{t-1} + u_t \tag{21.3.9}$$

where δ is known as the **drift parameter.** The name drift comes from the fact that if we write the preceding equation as

$$Y_t - Y_{t-1} = \Delta Y_t = \delta + u_t \tag{21.3.10}$$

it shows that Y_t drifts upward or downward, depending on δ being positive or negative. Note that model (21.3.9) is also an AR(1) model.

Following the procedure discussed for random walk without drift, it can be shown that for the random walk with drift model (21.3.9),

$$E(Y_t) = Y_0 + t \cdot \delta \tag{21.3.11}$$

$$\text{var}\,(Y_t) = t\sigma^2 \tag{21.3.12}$$

As you can see, for RWM with drift the mean as well as the variance increases over time, again violating the conditions of (weak) stationarity. In short, RWM, with or without drift, is a nonstationary stochastic process.

To give a glimpse of the random walk with and without drift, we conducted two simulations as follows:

$$Y_t = Y_0 + u_t \tag{21.3.13}$$

where u_t are white noise error terms such that each $u_t \sim N(0, 1)$; that is, each u_t follows the standard normal distribution. From a random number generator, we obtained 500 values of u and generated Y_t as shown in Eq. (21.3.13). We assumed $Y_0 = 0$. Thus, Eq. (21.3.13) is an RWM without drift.

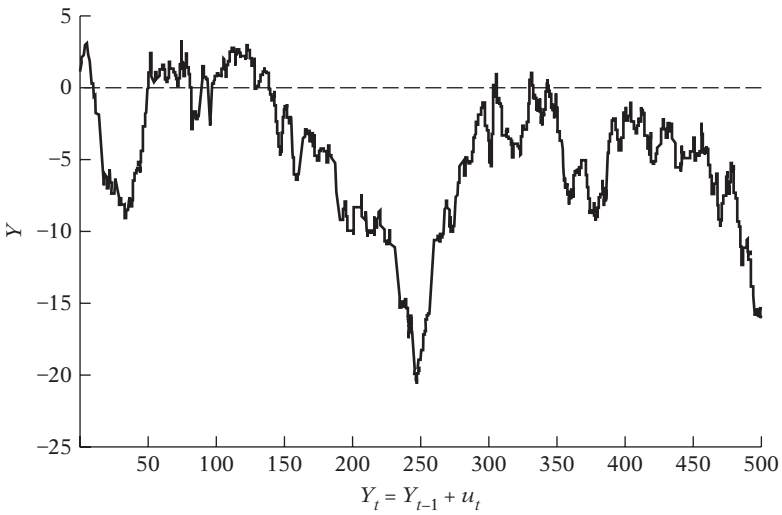
Now consider

$$Y_t = \delta + Y_0 + u_t \tag{21.3.14}$$

which is RWM with drift. We assumed u_t and Y_0 as in Eq. (21.3.13) and assumed that $\delta = 2$.

The graphs of models (21.3.13) and (21.3.14), respectively, are in Figures 21.3 and 21.4. The reader can compare these two diagrams in light of our discussion of the RWM with and without drift.

FIGURE 21.3
A random walk without drift.



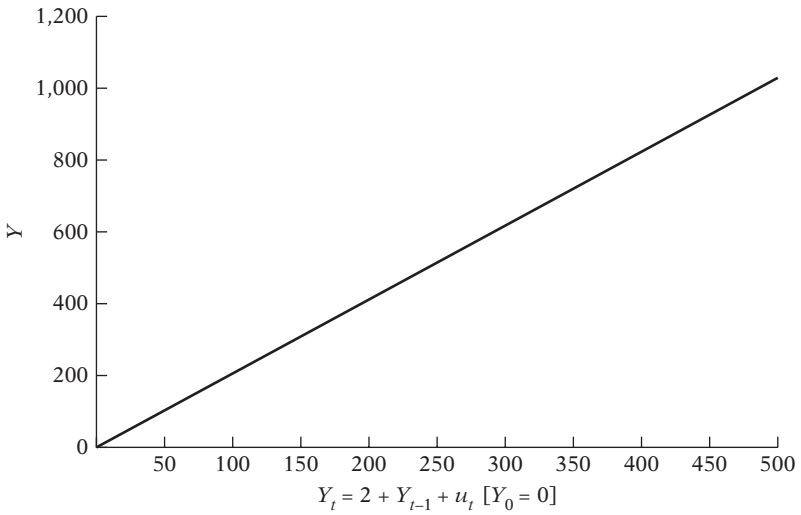


FIGURE 21.4
A random walk with drift.

The random walk model is an example of what is known in the literature as a **unit root process.** Since this term has gained tremendous currency in the time series literature, we next explain what a unit root process is.

21.4 Unit Root Stochastic Process

Let us write the RWM (21.3.4) as:

$$Y_t = \rho Y_{t-1} + u_t \qquad -1 \leq \rho \leq 1 \qquad \textbf{(21.4.1)}$$

This model resembles the Markov first-order autoregressive model that we discussed in the chapter on autocorrelation. If $\rho = 1$, Eq. (21.4.1) becomes a RWM (without drift). If ρ is in fact 1, we face what is known as the **unit root problem,** that is, a situation of nonstationarity; we already know that in this case the variance of Y_t is not stationary. The name unit root is due to the fact that $\rho = 1$.[11] *Thus the terms nonstationarity, random walk, unit root, and stochastic trend can be treated synonymously.*

If, however, $|\rho| < 1$, that is if the absolute value of ρ is less than one, then it can be shown that the time series Y_t is stationary in the sense we have defined it.[12]

In practice, then, it is important to find out if a time series possesses a unit root.[13] In Section 21.9 we will discuss several tests of unit root, that is, several tests of stationarity. In that section we will also determine whether the time series depicted in Figures 21.1 and 21.2 are stationary. Perhaps the reader might suspect that they are not. But we shall see.

[11]A technical point: If $\rho = 1$, we can write Eq. (21.4.1) as $Y_t - Y_{t-1} = u_t$. Now using the **lag operator** L so that $LY_t = Y_{t-1}$, $L^2Y_t = Y_{t-2}$, and so on, we can write Eq. (21.4.1) as $(1 - L)Y_t = u_t$. The term unit root refers to the root of the polynomial in the lag operator. If you set $(1 - L) = 0$, we obtain, $L = 1$, hence the name unit root.

[12]If in Eq. (21.4.1) it is assumed that the initial value of Y $(= Y_0)$ is zero, $|\rho| < 1$, and u_t is white noise and distributed normally with zero mean and unit variance, then it follows that $E(Y_t) = 0$ and var $(Y_t) = 1/(1 - \rho^2)$. Since both these are constants, by the definition of weak stationarity, Y_t is stationary. On the other hand, as we saw before, if $\rho = 1$, Y_t is a random walk or nonstationary.

[13]A time series may contain more than one unit root. But we will discuss this situation later in the chapter.

21.5 Trend Stationary (TS) and Difference Stationary (DS) Stochastic Processes

The distinction between stationary and nonstationary stochastic processes (or time series) has a crucial bearing on whether the trend (the slow long-run evolution of the time series under consideration) observed in the constructed time series in Figures 21.3 and 21.4 or in the actual economic time series of Figures 21.1 and 21.2 is **deterministic** or **stochastic.** Broadly speaking, if the trend in a time series is a deterministic function of time, such as time, time-squared etc., we call it a deterministic trend, whereas if it is not predictable, we call it a stochastic trend. To make the definition more formal, consider the following model of the time series Y_t.

$$Y_t = \beta_1 + \beta_2 t + \beta_3 Y_{t-1} + u_t \tag{21.5.1}$$

where u_t is a white noise error term and where t is time measured chronologically. Now we have the following possibilities:

Pure random walk: If in Eq. (21.5.1) $\beta_1 = 0, \beta_2 = 0, \beta_3 = 1,$ we get

$$Y_t = Y_{t-1} + u_t \tag{21.5.2}$$

which is nothing but a RWM without drift and is therefore nonstationary. But note that, if we write Eq. (21.5.2) as

$$\Delta Y_t = (Y_t - Y_{t-1}) = u_t \tag{21.3.8}$$

it becomes stationary, as noted before. Hence, a RWM without drift is a **difference stationary process (DSP).**

Random walk with drift: If in Eq. (21.5.1) $\beta_1 \neq 0, \beta_2 = 0, \beta_3 = 1,$ we get

$$Y_t = \beta_1 + Y_{t-1} + u_t \tag{21.5.3}$$

which is a random walk with drift and is therefore nonstationary. If we write it as

$$(Y_t - Y_{t-1}) = \Delta Y_t = \beta_1 + u_t \tag{21.5.3a}$$

this means Y_t will exhibit a positive ($\beta_1 > 0$) or negative ($\beta_1 < 0$) trend (see Figure 21.4). Such a trend is called a **stochastic trend.** Equation (21.5.3*a*) is a DSP process because the nonstationarity in Y_t can be eliminated by taking first differences of the time series. Remember that u_t in Eq. (21.5.3*a*) is a white noise error term.

Deterministic trend: If in Eq. (21.5.1), $\beta_1 \neq 0, \beta_2 \neq 0, \beta_3 = 0,$ we obtain

$$Y_t = \beta_1 + \beta_2 t + u_t \tag{21.5.4}$$

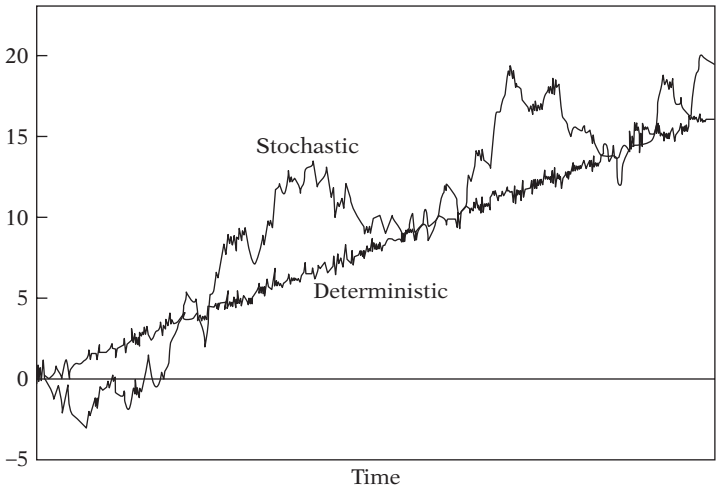
which is called a **trend stationary process (TSP).** Although the mean of Y_t is $\beta_1 + \beta_2 t,$ which is not constant, its variance ($= \sigma^2$) is. Once the values of β_1 and β_2 are known, the mean can be forecast perfectly. Therefore, if we subtract the mean of Y_t from Y_t, the resulting series will be stationary, hence the name **trend stationary.** This procedure of removing the (deterministic) trend is called **detrending.**

Random walk with drift and deterministic trend: If in Eq. (21.5.1), $\beta_1 \neq 0,$ $\beta_2 \neq 0, \beta_3 = 1,$ we obtain:

$$Y_t = \beta_1 + \beta_2 t + Y_{t-1} + u_t \tag{21.5.5}$$

FIGURE 21.5
Deterministic versus stochastic trend.

Source: Charemza et al., op. cit., p. 91.



in which case we have a random walk with drift and a deterministic trend, which can be seen if we write this equation as

$$\Delta Y_t = \beta_1 + \beta_2 t + u_t \qquad \textbf{(21.5.5}a\textbf{)}$$

which means that Y_t is nonstationary.

Deterministic trend with stationary AR(1) component: If in Eq. (21.5.1) $\beta_1 \neq 0$, $\beta_2 \neq 0$, $\beta_3 < 1$, then we get

$$Y_t = \beta_1 + \beta_2 t + \beta_3 Y_{t-1} + u_t \qquad \textbf{(21.5.6)}$$

which is stationary around the deterministic trend.

To see the difference between stochastic and deterministic trends, consider Figure 21.5.[14] The series named stochastic in this figure is generated by an RWM with drift: $Y_t = 0.5 + Y_{t-1} + u_t$, where 500 values of u_t were generated from a standard normal distribution and where the initial value of Y was set at 1. The series named deterministic is generated as follows: $Y_t = 0.5t + u_t$, where u_t were generated as above and where t is time measured chronologically.

As you can see from Figure 21.5, in the case of the deterministic trend, the deviations from the trend line (which represents the nonstationary mean) are purely random and they die out quickly; they do not contribute to the long-run development of the time series, which is determined by the trend component $0.5t$. In the case of the stochastic trend, on the other hand, the random component u_t affects the long-run course of the series Y_t.

21.6 Integrated Stochastic Processes

The random walk model is but a specific case of a more general class of stochastic processes known as **integrated processes.** Recall that the RWM without drift is nonstationary, but its first difference, as shown in Eq. (21.3.8), is stationary. Therefore, we call the RWM without drift **integrated of order 1,** denoted as $I(1)$. Similarly, if a time series has to be differenced twice (i.e., take the first difference of the first differences) to make it stationary, we call such a time series **integrated of order 2.**[15] In general, if a (nonstationary) time

[14]The following discussion is based on Wojciech W. Charemza et al., op. cit., pp. 89–91.

[15]For example if Y_t is $I(2)$, then $\Delta\Delta Y_t = \Delta(Y_t - Y_{t-1}) = \Delta Y_t - \Delta Y_{t-1} = Y_t - 2Y_{t-1} + Y_{t-2}$ will become stationary. But note that $\Delta\Delta Y_t = \Delta^2 Y_t \neq Y_t - Y_{t-2}$.

series has to be differenced d times to make it stationary, that time series is said to be **integrated of order *d*.** A time series Y_t integrated of order d is denoted as $Y_t \sim I(d)$. If a time series Y_t is stationary to begin with (i.e., it does not require any differencing), it is said to be integrated of order zero, denoted by $Y_t \sim I(0)$. Thus, we will use the terms "stationary time series" and "time series integrated of order zero" to mean the same thing.

Most economic time series are generally $I(1)$; that is, they generally become stationary only after taking their first differences. Are the time series shown in Figures 21.1 and 21.2 $I(1)$ or of higher order? We will examine them in Sections 21.8 and 21.9.

Properties of Integrated Series

The following properties of integrated time series may be noted: Let X_t, Y_t, and Z_t be three time series.

1. If $X_t \sim I(0)$ and $Y_t \sim I(1)$, then $Z_t = (X_t + Y_t) = I(1)$; that is, a linear combination or sum of stationary and nonstationary time series is nonstationary.
2. If $X_t \sim I(d)$, then $Z_t = (a + bX_t) = I(d)$, where a and b are constants. That is, a linear combination of an $I(d)$ series is also $I(d)$. Thus, if $X_t \sim I(0)$, then $Z_t = (a + bX_t) \sim I(0)$.
3. If $X_t \sim I(d_1)$ and $Y_t \sim I(d_2)$, then $Z_t = (aX_t + bY_t) \sim I(d_2)$, where $d_1 < d_2$.
4. If $X_t \sim I(d)$ and $Y_t \sim I(d)$, then $Z_t = (aX_t + bY_t) \sim I(d^*)$; d^* is generally equal to d, but in some cases $d^* < d$ (see the topic of cointegration in Section 21.11).

As you can see from the preceding statements, one has to pay careful attention in combining two or more time series that are integrated of different order.

To see why this is important, consider the two-variable regression model discussed in Chapter 3, namely, $Y_t = \beta_1 + \beta_2 X_t + u_t$. Under the classical OLS assumptions, we know that

$$\hat{\beta}_2 = \frac{\sum x_t y_t}{\sum x_t^2} \qquad \textbf{(21.6.1)}$$

where the small letters, as usual, indicate deviation from mean values. Suppose Y_t is $I(0)$, but X_t is $I(1)$; that is, the former is stationary and the latter is not. Since X_t is nonstationary, its variance will increase indefinitely, thus dominating the numerator term in Eq. (21.6.1) with the result that $\hat{\beta}_2$ will converge to zero asymptotically (i.e., in large samples) and it will not even have an asymptotic distribution.[16]

21.7 The Phenomenon of Spurious Regression

To see why stationary time series are so important, consider the following two random walk models:

$$Y_t = Y_{t-1} + u_t \qquad \textbf{(21.7.1)}$$

$$X_t = X_{t-1} + v_t \qquad \textbf{(21.7.2)}$$

where we generated 500 observations of u_t from $u_t \sim N(0, 1)$ and 500 observations of v_t from $v_t \sim N(0, 1)$ and assumed that the initial values of both Y and X were zero. We also assumed that u_t and v_t are serially uncorrelated as well as mutually uncorrelated. As you know by now, both these time series are nonstationary; that is, they are $I(1)$ or exhibit stochastic trends.

[16]This point is due to Maddala et al., op. cit., p. 26.

Suppose we regress Y_t on X_t. Since Y_t and X_t are uncorrelated $I(1)$ processes, the R^2 from the regression of Y on X should tend to zero; that is, there should not be any relationship between the two variables. But wait till you see the regression results:

Variable	Coefficient	Std. Error	t Statistic
C	-13.2556	0.6203	-21.36856
X	0.3376	0.0443	7.61223

$R^2 = 0.1044$ $d = 0.0121$

As you can see, the coefficient of X is highly statistically significant, and, although the R^2 value is low, it is statistically significantly different from zero. From these results, you may be tempted to conclude that there is a significant statistical relationship between Y and X, whereas a priori there should be none. This is in a nutshell the **phenomenon of spurious** or **nonsense regression,** first discovered by Yule.[17] Yule showed that (spurious) correlation could persist in nonstationary time series even if the sample is very large. That there is something wrong in the preceding regression is suggested by the extremely low Durbin–Watson d value, which suggests very strong first-order autocorrelation. According to Granger and Newbold, *an $R^2 > d$ is a good rule of thumb to suspect that the estimated regression is spurious,* as in the example above. It may be added that the R^2 and the t statistic from such a spurious regression are misleading, and the t statistics are not distributed as (Student's) t distribution and, therefore, cannot be used for testing hypotheses about the parameters.

That the regression results presented above are meaningless can be easily seen from regressing the first differences of $Y_t\ (= \Delta Y_t)$ on the first differences of $X_t\ (= \Delta X_t)$; remember that although Y_t and X_t are nonstationary, their first differences are stationary. In such a regression you will find that R^2 is practically zero, as it should be, and the Durbin–Watson d is about 2. In Exercise 21.24 you are asked to run this regression and verify the statement just made.

Although dramatic, this example is a strong reminder that one should be extremely wary of conducting regression analyses based on time series that exhibit stochastic trends. And one should therefore be extremely cautious in reading too much into the regression results based on $I(1)$ variables. For an example, see Exercise 21.26. To some extent, this is true of time series subject to deterministic trends, an example of which is given in Exercise 21.25.

21.8 Tests of Stationarity

By now the reader probably has a good idea about the nature of stationary stochastic processes and their importance. In practice we face two important questions: (1) How do we find out if a given time series is stationary? (2) If we find that a given time series is not stationary, is there a way that it can be made stationary? We take up the first question in this section and discuss the second question in Section 21.10.

Before we proceed, keep in mind that we are primarily concerned with weak, or covariance, stationarity.

Although there are several tests of stationarity, we discuss only those that are prominently discussed in the literature. In this section we discuss two tests: (1) graphical analysis and

[17]G. U. Yule, "Why Do We Sometimes Get Nonsense Correlations Between Time Series? A Study in Sampling and the Nature of Time Series," *Journal of the Royal Statistical Society,* vol. 89, 1926, pp. 1–64. For extensive Monte Carlo simulations on spurious regression see C. W. J. Granger and P. Newbold, "Spurious Regressions in Econometrics," *Journal of Econometrics,* vol. 2, 1974, pp. 111–120.

(2) the correlogram test. Because of the importance attached to it in the recent past, we discuss the *unit root test* in the next section. We illustrate these tests with appropriate examples.

1. Graphical Analysis

As noted earlier, before one pursues formal tests, it is always advisable to plot the time series under study, as we have done in Figures 21.1 and 21.2 for the U.S. economic time series data posted on the book's website. Such plots give an initial clue about the likely nature of the time series. Take, for instance, the GDP time series shown in Figure 21.1. You will see that over the period of study the log of GDP has been increasing, that is, showing an upward trend, suggesting perhaps that the mean of the log of GDP has been changing. This perhaps suggests that the log of the GDP series is not stationary. This is also more or less true of the other U.S. economic time series shown in Figure 21.2. Such an intuitive feel is the starting point of more formal tests of stationarity.

2. Autocorrelation Function (ACF) and Correlogram

One simple test of stationarity is based on the so-called **autocorrelation function (ACF).** The ACF at lag k, denoted by ρ_k, is defined as

$$\begin{aligned}\rho_k &= \frac{\gamma_k}{\gamma_0} \\ &= \frac{\text{covariance at lag } k}{\text{variance}}\end{aligned} \tag{21.8.1}$$

where covariance at lag k and variance are as defined before. Note that if $k = 0$, $\rho_0 = 1$ (why?)

Since both covariance and variance are measured in the same units of measurement, ρ_k is a *unitless,* or *pure, number.* It lies between -1 and $+1$, as any correlation coefficient does. If we plot ρ_k against k, the graph we obtain is known as the **population correlogram.**

Since in practice we only have a realization (i.e., sample) of a stochastic process, we can only compute the **sample autocorrelation function (SAFC),** $\hat{\rho}_k$. To compute this, we must first compute the **sample covariance** at lag k, $\hat{\gamma}_k$, and the **sample variance,** $\hat{\gamma}_0$, which are defined as:[18]

$$\hat{\gamma}_k = \frac{\sum(Y_t - \bar{Y})(Y_{t+k} - \bar{Y})}{n} \tag{21.8.2}$$

$$\hat{\gamma}_0 = \frac{\sum(Y_t - \bar{Y})^2}{n} \tag{21.8.3}$$

where n is the sample size and $\bar{Y}$ is the sample mean.

Therefore, the sample autocorrelation function at lag k is:

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0} \tag{21.8.4}$$

which is simply the ratio of sample covariance (at lag k) to sample variance. A plot of $\hat{\rho}_k$ against k is known as the **sample correlogram.**

How does a sample correlogram enable us to find out if a particular time series is stationary? For this purpose, let us first present the sample correlograms of a purely white noise

[18]Strictly speaking, we should divide the sample covariance at lag k by $(n - k)$ and the sample variance by $(n - 1)$ rather than by n (why?), where n is the sample size.

FIGURE 21.6
Correlogram of white noise error term u. AC = autocorrelation, PAC = partial autocorrelation (see Chapter 22), Q-Stat = Q statistic, Prob = probability.

Sample: 2 500
Included observations: 499

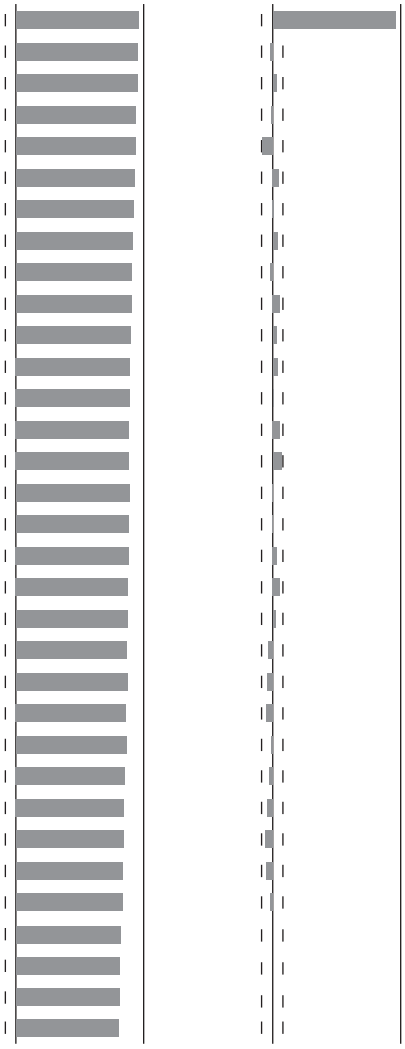
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	–0.022	–0.022	0.2335	0.629
		2	–0.019	–0.020	0.4247	0.809
		3	–0.009	–0.010	0.4640	0.927
		4	–0.031	–0.031	0.9372	0.919
		5	–0.070	–0.072	3.4186	0.636
		6	–0.008	–0.013	3.4493	0.751
		7	0.048	0.045	4.6411	0.704
		8	–0.069	–0.070	7.0385	0.532
		9	0.022	0.017	7.2956	0.606
		10	–0.004	–0.011	7.3059	0.696
		11	0.024	0.025	7.6102	0.748
		12	0.024	0.027	7.8993	0.793
		13	0.026	0.021	8.2502	0.827
		14	–0.047	–0.046	9.3726	0.806
		15	–0.037	–0.030	10.074	0.815
		16	–0.026	–0.031	10.429	0.843
		17	–0.029	–0.024	10.865	0.863
		18	–0.043	–0.050	11.807	0.857
		19	0.038	0.028	12.575	0.860
		20	0.099	0.093	17.739	0.605
		21	0.001	0.007	17.739	0.665
		22	0.065	0.060	19.923	0.588
		23	0.053	0.055	21.404	0.556
		24	–0.017	–0.004	21.553	0.606
		25	–0.024	–0.005	21.850	0.644
		26	–0.008	–0.008	21.885	0.695
		27	–0.036	–0.027	22.587	0.707
		28	0.053	0.072	24.068	0.678
		29	–0.004	–0.011	24.077	0.725
		30	–0.026	–0.025	24.445	0.752

random process and of a random walk process. Return to the driftless RWM (21.3.13). There we generated a sample of 500 error terms, the u's, from the standard normal distribution. The correlogram of these 500 purely random error terms is as shown in Figure 21.6; we have shown this correlogram up to 30 lags. We will comment shortly on how one chooses the lag length.

For the time being, just look at the column labeled AC, which is the sample autocorrelation function, and the first diagram on the left, labeled Autocorrelation. The solid vertical line in this diagram represents the zero axis; observations to the right of the line are positive values and those to the left of the line are negative values. As is very clear from this diagram, for a purely white noise process the autocorrelations at various lags hover around zero. *This is the picture of a correlogram of a stationary time series.* Thus, if the correlogram of an actual (economic) time series resembles the correlogram of a white noise time series, we can say that time series is probably stationary.

FIGURE 21.7
Correlogram of a random walk time series. See Figure 21.6 for definitions.

Sample: 2 500
Included observations: 499



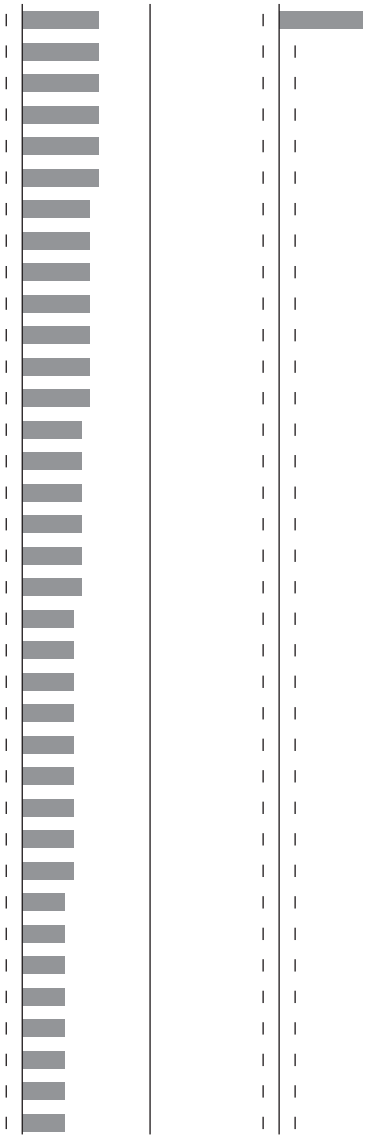
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.992	0.992	493.86	0.000
		2	0.984	0.000	980.68	0.000
		3	0.976	0.030	1461.1	0.000
		4	0.969	0.005	1935.1	0.000
		5	0.961	–0.059	2402.0	0.000
		6	0.953	0.050	2862.7	0.000
		7	0.946	0.004	3317.3	0.000
		8	0.939	0.040	3766.4	0.000
		9	0.932	–0.009	4210.1	0.000
		10	0.927	0.055	4649.1	0.000
		11	0.921	0.018	5083.9	0.000
		12	0.916	0.039	5514.9	0.000
		13	0.912	0.002	5942.4	0.000
		14	0.908	0.056	6367.0	0.000
		15	0.905	0.061	6789.8	0.000
		16	0.902	0.000	7210.6	0.000
		17	0.899	0.006	7629.4	0.000
		18	0.896	0.030	8046.7	0.000
		19	0.894	0.053	8463.1	0.000
		20	0.892	0.013	8878.7	0.000
		21	0.890	–0.041	9292.6	0.000
		22	0.886	–0.040	9704.1	0.000
		23	0.882	–0.044	10113.	0.000
		24	0.878	–0.012	10518.	0.000
		25	0.873	–0.023	10920.	0.000
		26	0.867	–0.041	11317.	0.000
		27	0.860	–0.055	11709.	0.000
		28	0.853	–0.045	12095.	0.000
		29	0.846	–0.010	12476.	0.000
		30	0.839	0.008	12851.	0.000
		31	0.832	–0.006	13221.	0.000
		32	0.825	0.003	13586.	0.000
		33	0.819	–0.006	13946.	0.000

Now look at the correlogram of a random walk series, as generated, say, by Eq. (21.3.13). The picture is as shown in Figure 21.7. The most striking feature of this correlogram is that the autocorrelation coefficients at various lags are very high even up to a lag of 33 quarters. As a matter of fact, if we consider lags of up to 60 quarters, the autocorrelation coefficients are quite high; the coefficient is about 0.7 at lag 60. Figure 21.7 is the typical correlogram of a nonstationary time series: The autocorrelation coefficient starts at a very high value and declines very slowly toward zero as the lag lengthens.

Now let us take a concrete economic example. Let us examine the correlogram of the LGDP time series plotted using the U.S. economic times series data posted on the book's website (see Section 21.1). The correlogram up to 36 lags is shown in Figure 21.8. The LGDP correlogram up to 36 lags also shows a pattern similar to the correlogram of the random walk

FIGURE 21.8
Correlogram of U.S. LGDP, 1947–I to 2007–IV. See Figure 21.6 for definitions.

Sample: 1947–I 2007–IV
Included observations: 244



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.977	0.977	235.73	0.000
		2	0.954	–0.009	461.43	0.000
		3	0.931	–0.010	677.31	0.000
		4	0.908	–0.006	883.67	0.000
		5	0.886	–0.003	1080.9	0.000
		6	0.864	–0.001	1269.3	0.000
		7	0.843	–0.006	1449.3	0.000
		8	0.822	–0.006	1621.0	0.000
		9	0.801	–0.010	1784.6	0.000
		10	0.780	–0.004	1940.6	0.000
		11	0.759	–0.007	2089.0	0.000
		12	0.738	–0.013	2230.0	0.000
		13	0.718	0.003	2364.1	0.000
		14	0.699	–0.005	2491.5	0.000
		15	0.679	–0.001	2612.4	0.000
		16	0.660	–0.004	2727.2	0.000
		17	0.642	–0.002	2836.2	0.000
		18	0.624	0.002	2939.6	0.000
		19	0.607	0.003	3037.8	0.000
		20	0.590	–0.003	3130.9	0.000
		21	0.573	–0.003	3219.3	0.000
		22	0.557	–0.003	3303.1	0.000
		23	0.541	–0.001	3382.5	0.000
		24	0.526	0.007	3457.9	0.000
		25	0.511	0.002	3529.4	0.000
		26	0.496	–0.005	3597.2	0.000
		27	0.482	–0.011	3661.4	0.000
		28	0.467	–0.009	3722.0	0.000
		29	0.453	–0.005	3779.2	0.000
		30	0.438	–0.006	3833.1	0.000
		31	0.424	–0.005	3883.9	0.000
		32	0.411	0.004	3931.6	0.000
		33	0.398	0.004	3976.7	0.000
		34	0.385	–0.001	4019.1	0.000
		35	0.373	–0.009	4058.9	0.000
		36	0.360	–0.010	4096.3	0.000

model in Figure 21.7. The autocorrelation coefficient starts at a very high value at lag 1 (0.977) and declines very slowly. Thus it seems that the LGDP time series is nonstationary. If you plot the correlograms of the other U.S. economic time series shown in Figures 21.1 and 21.2, you will also see a similar pattern, leading to the conclusion that all these time series are nonstationary; they may be nonstationary in mean or variance or both.

Two practical questions may be posed here. First, how do we choose the lag length to compute the ACF? Second, how do you decide whether a correlation coefficient at a certain lag is statistically significant? The answer follows.

The Choice of Lag Length

This is basically an empirical question. A rule of thumb is to compute ACF up to one-third to one-quarter the length of the time series. Since for our economic data we have 244 quarterly observations, by this rule lags of 61 to 81 quarters will do. To save space, we have only shown 36 lags in the ACF graph in Figure 21.8. The best practical advice is to start with sufficiently large lags and then reduce them by some statistical criterion, such as the *Akaike* or *Schwarz information criterion* that we discussed in Chapter 13. Alternatively, one can use the following statistical tests.

Statistical Significance of Autocorrelation Coefficients

Consider, for instance, the correlogram of the LGDP time series given in Figure 21.8. How do we decide whether the correlation coefficient of 0.780 at lag 10 (quarters) is statistically significant? The statistical significance of any $\hat{\rho}_k$ can be judged by its standard error. Bartlett has shown that if a time series is purely random, that is, it exhibits white noise (see Figure 21.6), the sample autocorrelation coefficients $\hat{\rho}_k$ are *approximately*[19]

$$\hat{\rho}_k \sim N(0, 1/n) \tag{21.8.5}$$

that is, in large samples the sample autocorrelation coefficients are normally distributed with zero mean and variance equal to one over the sample size. Since we have 244 observations, the variance is $1/244 \approx 0.0041$ and the standard error is $\sqrt{0.0041} \approx 0.0640$. Then following the properties of the standard normal distribution, the 95 percent confidence interval for any (population) ρ_k is:

$$\hat{\rho}_k \pm 1.96(0.0640) = \hat{\rho}_k \pm 0.1254 \tag{21.8.6}$$

In other words,

$$\text{Prob}\,(\hat{\rho}_k - 0.1254 \leq \rho_k \leq \hat{\rho}_k + 0.1254) = 0.95 \tag{21.8.7}$$

If the preceding interval includes the value of zero, we do not reject the hypothesis that the true ρ_k is zero, but if this interval does not include 0, we reject the hypothesis that the true ρ_k is zero. Applying this to the estimated value of $\hat{\rho}_{10} = 0.873$, the reader can verify that the 95 percent confidence interval for true ρ_{10} is (0.873 ± 0.1254) or $(0.7476, 0.9984)$.[20] Obviously, this interval does not include the value of zero, suggesting that we are 95 percent confident that the true ρ_{10} is significantly different from zero.[21] As you can check, even at lag 20 the estimated ρ_{20} is statistically significant at the 5 percent level.

Instead of testing the statistical significance of any individual autocorrelation coefficient, we can test the *joint hypothesis* that all the ρ_k up to certain lags are simultaneously equal to zero. This can be done by using the ***Q* statistic** developed by Box and Pierce, which is defined as[22]

$$Q = n \sum_{k=1}^{m} \hat{\rho}_k^2 \tag{21.8.8}$$

[19]M. S. Bartlett, "On the Theoretical Specification of Sampling Properties of Autocorrelated Time Series," *Journal of the Royal Statistical Society,* Series B, vol. 27, 1946, pp. 27–41.

[20]Our sample size of 244 observations is reasonably large to use the normal approximation.

[21]Alternatively, if you divide the estimated value of any ρ_k by the standard error of $(\sqrt{1/n})$, for sufficiently large n, you will obtain the standard Z value, whose probability can be easily obtained from the standard normal table. Thus for the estimated $\rho_{10} = 0.780$, the Z value is $0.780/0.1066 = 7.32$ (approx.). If the true ρ_{10} were in fact zero, the probability of obtaining a Z value of as much as 7.32 or greater is very small, thus rejecting the hypothesis that the true ρ_{10} is zero.

[22] G. E. P. Box and D. A. Pierce, "Distribution of Residual Autocorrelations in Autoregressive Integrated Moving Average Time Series Models," *Journal of the American Statistical Association*, vol. 65, 1970, pp. 1509–1526.

where n = sample size and m = lag length. The Q statistic is often used as a test of whether a time series is white noise. In large samples, it is *approximately* distributed as the chi-square distribution with m df. In an application, if the computed Q exceeds the critical Q value from the chi-square distribution at the chosen level of significance, one can reject the null hypothesis that all the (true) ρ_k are zero; at least some of them must be nonzero.

A variant of the Box–Pierce Q statistic is the **Ljung–Box (LB) statistic,** which is defined as[23]

$$\text{LB} = n(n+2)\sum_{k=1}^{m}\left(\frac{\hat{\rho}_k^2}{n-k}\right) \sim \chi^2 m \qquad \textbf{(21.8.9)}$$

Although in large samples both Q and LB statistics follow the chi-square distribution with m df, the LB statistic has been found to have better (more powerful, in the statistical sense) small-sample properties than the Q statistic.[24]

Returning to the LGDP example given in Figure 21.8, the value of the Q statistic up to lag 36 is about 4096. The probability of obtaining such a Q value under the null hypothesis that the sum of 36 squared estimated autocorrelation coefficients is zero is practically zero, as the last column of that figures shows. Therefore, the conclusion is that the LGDP time series is probably nonstationary, therefore reinforcing our hunch from Figure 21.1 that the LGDP series may be nonstationary. In Exercise 21.16 you are asked to confirm that the other four U.S. economic time series are also nonstationary.

21.9 The Unit Root Test

A test of stationarity (or nonstationarity) that has become widely popular over the past several years is the **unit root test.** We will first explain it, then illustrate it, and then consider some of its limitations.

The starting point is the unit root (stochastic) process that we discussed in Section 21.4. We start with

$$Y_t = \rho Y_{t-1} + u_t \qquad -1 \leq \rho \leq 1 \qquad \textbf{(21.4.1)}$$

where u_t is a white noise error term.

We know that if $\rho = 1$, that is, in the case of the unit root, Eq. (21.4.1) becomes a random walk model without drift, which we know is a nonstationary stochastic process. Therefore, why not simply regress Y_t on its (one-period) lagged value Y_{t-1} and find out if the estimated ρ is statistically equal to 1? If it is, then Y_t is nonstationary. This is the general idea behind the unit root test of stationarity.

However, we cannot estimate Eq. (21.4.1) by OLS and test the hypothesis that $\rho = 1$ by the usual t test because that test is severely biased in the case of a unit root. Therefore, we manipulate Eq. (21.4.1) as follows: Subtract Y_{t-1} from both sides of Eq. (21.4.1) to obtain:

$$\begin{aligned} Y_t - Y_{t-1} &= \rho Y_{t-1} - Y_{t-1} + u_t \\ &= (\rho - 1)Y_{t-1} + u_t \end{aligned} \qquad \textbf{(21.9.1)}$$

which can be alternatively written as:

$$\Delta Y_t = \delta Y_{t-1} + u_t \qquad \textbf{(21.9.2)}$$

where $\delta = (\rho - 1)$ and Δ, as usual, is the first difference operator.

[23]G. M. Ljung and G. E. P. Box, "On a Measure of Lack of Fit in Time Series Models," *Biometrika,* vol. 66, 1978, pp. 66–72.

[24]The Q and LB statistics may not be appropriate in every case. For a critique, see Maddala et al., op. cit., p. 19.

In practice, therefore, instead of estimating Eq. (21.4.1), we estimate Eq. (21.9.2) and test the (null) hypothesis that $\delta = 0$, the alternative hypothesis being that $\delta < 0$ (see footnote 25). If $\delta = 0$, then $\rho = 1$, that is we have a unit root, meaning the time series under consideration is nonstationary.

Before we proceed to estimate Eq. (21.9.2), it may be noted that if $\delta = 0$, Eq. (21.9.2) will become

$$\Delta Y_t = (Y_t - Y_{t-1}) = u_t \tag{21.9.3}$$

Since u_t is a white noise error term, it is stationary, which means that the first differences of a random walk time series are stationary, a point we have already made before.

Now let us turn to the estimation of Eq. (21.9.2). This is simple enough; all we have to do is to take the first differences of Y_t and regress them on Y_{t-1} and see if the estimated slope coefficient in this regression ($= \hat{\delta}$) is zero or not. If it is zero, we conclude that Y_t is nonstationary. But if it is negative, we conclude that Y_t is stationary.[25] The only question is which test we use to find out if the estimated coefficient of Y_{t-1} in Eq. (21.9.2) is zero or not. You might be tempted to say, why not use the usual t test? Unfortunately, under the null hypothesis that $\delta = 0$ (i.e., $\rho = 1$), the t value of the estimated coefficient of Y_{t-1} does not follow the t distribution even in large samples; that is, it does not have an asymptotic normal distribution.

What is the alternative? Dickey and Fuller have shown that under the null hypothesis that $\delta = 0$, the estimated t value of the coefficient of Y_{t-1} in Eq. (21.9.2) follows the **τ (tau) statistic.**[26] These authors have computed the critical values of the *tau statistic* on the basis of Monte Carlo simulations. A sample of these critical values is given in **Appendix D,** Table D.7. The table is limited, but MacKinnon has prepared more extensive tables, which are now incorporated in several econometric packages.[27] In the literature the **tau statistic or test** is known as the **Dickey–Fuller (DF) test,** in honor of its discoverers. Interestingly, if the hypothesis that $\delta = 0$ is rejected (i.e., the time series is stationary), we can use the usual (Student's) t test. Keep in mind that the Dickey–Fuller test is one-sided because the alternative hypothesis is that $\delta < 0$ (or $\rho < 1$).

The actual procedure of implementing the DF test involves several decisions. In discussing the nature of the unit root process in Sections 21.4 and 21.5, we noted that a random walk process may have no drift, or it may have drift, or it may have both deterministic and stochastic trends. To allow for the various possibilities, the DF test is estimated in three different forms, that is, under three different null hypotheses.

Y_t is a random walk: $\Delta Y_t = \delta Y_{t-1} + u_t$ **(21.9.2)**

Y_t is a random walk with drift: $\Delta Y_t = \beta_1 + \delta Y_{t-1} + u_t$ **(21.9.4)**

Y_t is a random walk with drift around a deterministic trend: $\Delta Y_t = \beta_1 + \beta_2 t + \delta Y_{t-1} + u_t$ **(21.9.5)**

[25]Since $\delta = (\rho - 1)$, for stationarity ρ must be less than one. For this to happen δ must be negative.

[26]D. A. Dickey and W. A. Fuller, "Distribution of the Estimators for Autoregressive Time Series with a Unit Root," *Journal of the American Statistical Association,* vol. 74, 1979, pp. 427–431. See also W. A. Fuller, *Introduction to Statistical Time Series,* John Wiley & Sons, New York, 1976.

[27]J. G. MacKinnon, "Critical Values of Cointegration Tests," in R. E. Engle and C. W. J. Granger, eds., *Long-Run Economic Relationships: Readings in Cointegration,* Chapter 13, Oxford University Press, New York, 1991.

where t is the time or trend variable. In each case the hypotheses are:

Null hypothesis: $H_0 : \delta = 0$ (i.e., there is a unit root or the time series is nonstationary, or it has a stochastic trend).

Alternative hypothesis: $H_1 : \delta < 0$ (i.e., the time series is stationary, possibly around a deterministic trend).[28]

If the null hypothesis is rejected, it means either (1) Y_t is stationary with zero mean, in the case of Eq. (21.9.2), or (2) Y_t is stationary with nonzero mean, in the case of Eq. (21.9.4). In the case of Eq. (21.9.5), we can test for $\delta < 0$ (i.e., no stochastic trend) and $\alpha \neq 0$ (i.e., the existence of a deterministic trend) simultaneously, using the F test, but using the critical values tabulated by Dickey and Fuller. It may be noted that a time series may contain both a stochastic and a deterministic trend.

It is extremely important to note that the critical values of the tau test to test the hypothesis that $\delta = 0$ are different for each of the preceding three specifications of the DF test, which can be seen clearly from **Appendix D,** Table D.7. Moreover, if, say, specification (21.9.4) is correct, but we estimate Eq. (21.9.2), we will be committing a specification error, whose consequences we already know from Chapter 13. The same is true if we estimate Eq. (21.9.4) rather than the true Eq. (21.9.5). Of course, there is no way of knowing which specification is correct to begin with. Some trial and error is inevitable, data mining notwithstanding.

The actual estimation procedure is as follows: Estimate Eq. (21.9.2), or Eq. (21.9.3), or Eq. (21.9.4) by OLS; divide the estimated coefficient of Y_{t-1} in each case by its standard error to compute the (τ) tau statistic; and refer to the DF tables (or any statistical package). If the computed absolute value of the tau statistic ($|\tau|$) exceeds the *absolute* DF or MacKinnon critical tau values, we reject the hypothesis that $\delta = 0$, in which case the time series is stationary. On the other hand, if the computed $|\tau|$ does not exceed the absolute critical tau value, we do not reject the null hypothesis, in which case the time series is nonstationary. Make sure that you use the appropriate critical τ values. In most applications the tau value will be negative. Therefore, alternatively we can say that if the computed (negative) tau value is smaller than (i.e., more negative than) the critical tau value, we reject the null hypothesis (i.e., the time series is stationary) otherwise, we do not reject it (i.e., the time series is nonstationary).

Let us return to the U.S. GDP time series. For this series, the results of the three regressions (21.9.2), (21.9.4), and (21.9.5) are as follows: The dependent variable in each case is $\Delta Y_t = \Delta \text{LGDP}_t$, where LGDP is the logarithm of real GDP.

$$\widehat{\Delta \text{LGDP}}_t = 0.000968 \text{LGDP}_{t-1} \tag{21.9.6}$$
$$t = (12.9270) \qquad R^2 = 0.0147 \qquad d = 1.3194$$

$$\widehat{\Delta \text{LGDP}}_t = 0.0221 - 0.00165 \text{LGDP}_{t-1} \tag{21.9.7}$$
$$t = (2.4342) \quad (-1.5294) \qquad R^2 = 0.0096 \qquad d = 1.3484$$

$$\widehat{\Delta \text{LGDP}}_t = 0.2092 + 0.0002t - 0.0269 \text{LGDP}_{t-1} \tag{21.9.8}$$
$$t = (1.8991) \quad (1.7040) \quad (-1.8102)$$
$$R^2 = 0.0215 \qquad d = 1.3308$$

[28]We rule out the possibility that $\delta > 0$, because in that case $\rho > 1$, in which case the underlying time series will be explosive.

Our primary interest in all these regressions is in the $t(=\tau)$ value of the LGDP_{t-1} coefficient. If you look at Table D.7 in **Appendix D,** you will see that the 5 percent critical tau values for sample size 250 (the closest number to our sample of 244 observations) are -1.95 (no intercept, no trend), -2.88 (intercept but no trend), and -3.43 (intercept as well as trend). *EViews* and other statistical packages provide critical values for the sample size used in the analysis.

Before we examine the results, we have to decide which of the three models may be appropriate. We should rule out model (21.9.6) because the coefficient of LGDP_{t-1}, which is equal to δ is positive. But since $\delta = (\rho - 1)$, a positive δ would imply that $\rho > 1$. Although a theoretical possibility, we rule this out because in this case the LGDP time series would be explosive.[29] That leaves us with models (21.9.7) and (21.9.8). In both cases the estimated δ coefficient is negative, implying that the estimated ρ is less than 1. For these two models, the estimated ρ values are 0.9984 and 0.9731, respectively. The only question now is if these values are statistically significantly below 1 for us to declare that the GDP time series is stationary.

For model (21.9.7) the estimated τ value is -1.5294, whereas the 5 percent critical τ value, as noted above, is -2.88. Since, in *absolute* terms, the former is smaller than the latter, our conclusion is that the LGDP time series is not stationary.[30]

The story is the same for model (21.9.8). The computed τ value of -1.8102, in absolute terms, is smaller than the 5 percent critical value of -3.43.

Therefore, on the basis of graphical analysis, the correlogram, and the Dickey–Fuller test, the conclusion is that for the quarterly periods of 1947 to 2007, the U.S. LGDP time series was nonstationary; i.e., it contained a unit root, or it had a stochastic trend.

The Augmented Dickey–Fuller (ADF) Test

In conducting the DF test as in Eqs. (21.9.2), (21.9.4), and (21.9.5), it was assumed that the error term u_t was uncorrelated. But in case the u_t are correlated, Dickey and Fuller have developed another test, known as the **augmented Dickey–Fuller (ADF) test.** This test is conducted by "augmenting" the preceding three equations by adding the lagged values of the dependent variable ΔY_t. To be specific, suppose we use Eq. (21.9.5). The ADF test here consists of estimating the following regression:

$$\Delta Y_t = \beta_1 + \beta_2 t + \delta Y_{t-1} + \sum_{i=1}^{m} \alpha_i \Delta Y_{t-i} + \varepsilon_t \qquad \textbf{(21.9.9)}$$

where ε_t is a pure white noise error term and where $\Delta Y_{t-1} = (Y_{t-1} - Y_{t-2})$, $\Delta Y_{t-2} = (Y_{t-2} - Y_{t-3})$, etc. The number of lagged difference terms to include is often determined empirically, the idea being to include enough terms so that the error term in Eq. (21.9.9) is serially uncorrelated, so that we can obtain an unbiased estimate of δ, the coefficient of lagged Y_{t-1}. *EViews 6* has an option that automatically selects the lag length based on Akaike, Schwarz, and other information criteria. In ADF we still test whether $\delta = 0$ and the ADF test follows the same asymptotic distribution as the DF statistic, so the same critical values can be used.

To give a glimpse of this procedure, we estimated Eq. (21.9.9) for the LGDP series. Since we have quarterly data, we decided to use four lags. The results of the ADF regression are as follows:[31]

[29]More technically, since Eq. (21.9.2) is a first-order difference equation, the so-called stability condition requires that $|\rho| < 1$.

[30]Another way of stating this is that the computed τ value should be more negative than the critical τ value, which is not the case here. Hence the conclusion stays. Since in general δ is expected to be negative, the estimated τ statistic will have a negative sign. Therefore, a large negative τ value is generally an indication of stationarity.

[31]Higher-order lagged differences were considered but they were insignificant.

$$\widehat{\Delta \text{LGDP}}_t = 0.2677 + 0.0003t - 0.0352\text{LGDP}_{t-1} + 0.2990\Delta\text{LGDP}_{t-1} + 0.1451\Delta\text{LGDP}_{t-2} - 0.0621\Delta\text{LGDP}_{t-3} - 0.0876\Delta\text{LGDP}_t$$

$$t = (2.4130)\ \ (2.2561)\ \ (-2.3443)\qquad (4.6255)\qquad (2.1575)\qquad (-0.9205)\qquad (-1.3438)$$

$$R^2 = 0.1617 \qquad d = 2.0075 \qquad \textbf{(21.9.10)}$$

The $t\ (=\tau)$ value of the lagged LGDP_{t-1} coefficient $(=\delta)$ is -2.3443, which in absolute terms is much less than even the 10 percent critical τ value of -3.1378, again suggesting that even after taking care of possible autocorrelation in the error term, the LGDP series is nonstationary. (*Note:* The @trend command in *EViews* automatically generates the time or trend variable.)

Could this be the result of our choosing only four lagged values of ΔLGDP? We used the Schwarz criterion using 14 lagged values of ΔLGDP, which gave the tau value δ of -1.8102. Even then, this tau value was not significant at the 10 percent level (the critical tau value at this level was -3.1376). It seems logged GDP is nonstationary.

Testing the Significance of More than One Coefficient: The *F* Test

Suppose we estimate model (21.9.5) and test the hypothesis that $\beta_1 = \beta_2 = 0$, that is, the model is RWM without drift and trend. To test this joint hypothesis, we can use the *restricted F* test discussed in Chapter 8. That is, we estimate Eq. (21.9.5) (the unrestricted regression) and then estimate Eq. (21.9.5) again, dropping the intercept and trend. Then we use the restricted F test as shown in Eq. (8.6.9), except that we cannot use the conventional F table to get the critical F values. As they did with the τ statistic, Dickey and Fuller have developed critical F values for this situation, a sample of which is given in **Appendix D,** Table D.7. An example is presented in Exercise 21.27.

The Phillips–Perron (PP) Unit Root Tests[32]

An important assumption of the DF test is that the error terms u_t are independently and identically distributed. The ADF test adjusts the DF test to take care of possible serial correlation in the error terms by adding the lagged difference terms of the regressand. Phillips and Perron use *nonparametric statistical methods* to take care of the serial correlation in the error terms without adding lagged difference terms. Since the asymptotic distribution of the PP test is the same as the ADF test statistic, we will not pursue this topic here.

Testing for Structural Changes

The macroeconomic data introduced in Section 21.1 (see the book's website for the actual data) are for the period 1947–2007, a period of 61 years. In this period the U.S. economy experienced several business cycles of varying durations. Business cycles are marked by periods of recessions and periods of expansions. It is quite likely that one business cycle is different from another, which may reflect **structural breaks** or **structural changes** in the economy.

For instance, take the first oil embargo in 1973. It quadrupled oil prices. Prices again increased substantially after the second oil embargo in 1979. Naturally, these shocks will affect economic behavior. Therefore, if we were to regress personal consumption expenditure (PCE) on disposable personal income (DPI), the intercept, the slope, or both are likely to change from one business cycle to another (recall the Chow test of structural breaks). This is what is meant by structural changes.

[32]P. C. B. Phillips and P. Perron, "Testing for a Unit Root in Time Series Regression," *Biometrika,* vol. 75, 1988, pp. 335–346. The PP test is now included in several software packages.

Perron, for instance, has argued that the standard tests of the unit root hypothesis may not be reliable in the presence of structural changes.[33] There are ways to test for structural changes and to account for them, the simplest involving the use of dummy variables. But a discussion of the various tests of structural breaks will take us far afield and is best left for the references.[34] However, see Exercise 21.28.

A Critique of the Unit Root Tests[35]

We have discussed several unit root tests and there are several more. The question is: Why are there so many unit root tests? The answer lies in the **size** and **power** of these tests. By size of a test we mean the level of significance (i.e., the probability of committing a Type I error) and by power of a test we mean the probability of rejecting the null hypothesis when it is false. The power of a test is calculated by subtracting the probability of a Type II error from 1; Type II error is the probability of accepting a false null hypothesis. The maximum power is 1. Most unit root tests are based on the null hypothesis that the time series under consideration has a unit root; that is, it is nonstationary. The alternative hypothesis is that the time series is stationary.

Size of Test

You will recall from Chapter 13 the distinction we made between the nominal and the true levels of significance. The DF test is sensitive to the way it is conducted. Remember that we discussed three varieties of the DF test: (1) a pure random walk, (2) a random walk with drift, and (3) a random walk with drift and trend. If, for example, the true model is (1) but we estimate (2), and conclude that, say, on the 5 percent level that the time series is stationary, this conclusion may be wrong because the true level of significance in this case is much larger than 5 percent.[36] The size distortion could also result from excluding moving average (MA) components from the model (on moving average, see Chapter 22).

Power of Test

Most tests of the DF type have low power; that is, they tend to accept the null of unit root more frequently than is warranted. That is, these tests may find a unit root even when none exists. There are several reasons for this. *First,* the power depends on the (time) *span* of the data more than the mere size of the sample. For a given sample size n, the power is greater when the span is large. Thus, a unit root test(s) based on 30 observations over a span of 30 years may have more power than one based on, say, 100 observations over a span of 100 days. *Second,* if $\rho \approx 1$ but not exactly 1, the unit root test may declare such a time series nonstationary. *Third,* these types of tests assume a single unit root; that is, they assume that the given time series is $I(1)$. But if a time series is integrated of order higher than 1, say, $I(2)$, there will be more than one unit root. In the latter case one may use the **Dickey–Pantula test.**[37] *Fourth,* if there are structural breaks in a time series (see the chapter on dummy variables) due to, say, the OPEC oil embargoes, the unit root tests may not catch them.

In applying the unit root tests one should therefore keep in mind the limitations of the tests. Of course, there have been modifications of these tests by Perron and Ng, Elliot,

[33]P. Perron, "The Great Crash, the Oil Price Shock and the Unit Root Hypothesis," *Econometrica,* vol. 57, 1989, pp. 1361–1401.

[34]For an accessible discussion, see James H. Stock and Mark W. Watson, *Introduction to Econometrics,* 2d ed., Pearson/Addison-Wesley, Boston, 2007, pp. 565–571. For a more thorough discussion, see G. S. Maddala and In-Moo Kim, *Unit Roots, Cointegration, and Structural Change,* Cambridge University Press, New York, 1998.

[35]For detailed discussion, see Terrence C. Mills, op. cit., pp. 87–88.

[36]For a Monte Carlo experiment about this, see Charemza et al., op. cit., p. 114.

[37]D. A. Dickey and S. Pantula, "Determining the Order of Differencing in Autoregressive Processes," *Journal of Business and Economic Statistics,* vol. 5, 1987, pp. 455–461.

Rothenberg and Stock, Fuller, and Leybounre.[38] Because of this, Maddala and Kim advocate that the traditional DF, ADF, and PP tests should be discarded. As econometric software packages incorporate the new tests, that may very well happen. But it should be added that as yet there is no uniformly powerful test of the unit root hypothesis.

21.10 Transforming Nonstationary Time Series

Now that we know the problems associated with nonstationary time series, the practical question is what to do. To avoid the spurious regression problem that may arise from regressing a nonstationary time series on one or more nonstationary time series, we have to transform nonstationary time series to make them stationary. The transformation method depends on whether the time series are difference stationary (DSP) or trend stationary (TSP). We consider each of these methods in turn.

Difference-Stationary Processes

If a time series has a unit root, the first differences of such time series are stationary.[39] Therefore, the solution here is to take the first differences of the time series.

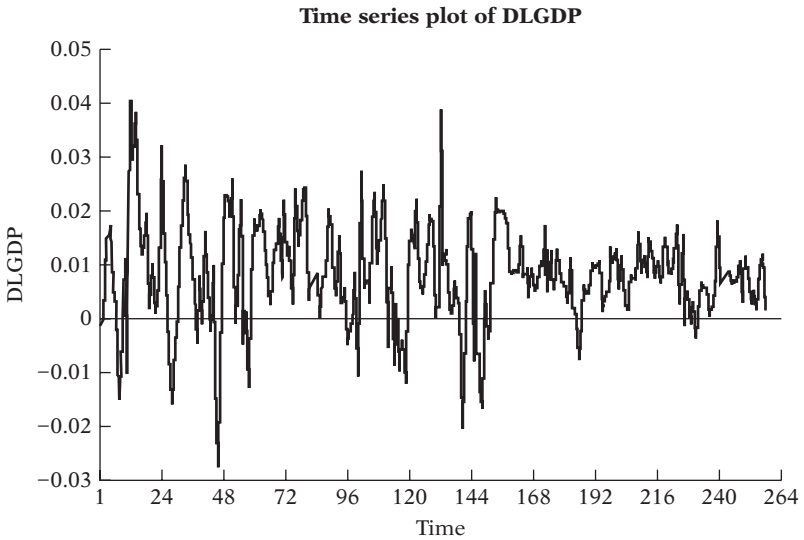
Returning to our U.S. LGDP time series, we have already seen that it has a unit root. Let us now see what happens if we take the first differences of the LGDP series.

Let $\Delta\text{LGDP}_t = (\text{LGDP}_t - \text{LGDP}_{t-1})$. For convenience, let $D_t = \Delta\text{LGDP}_t$. Now consider the following regression:

$$\begin{aligned} \widehat{\Delta D}_t &= \ 0.00557 - \quad 0.6711 D_{t-1} \\ t &= (7.1407) \quad (-11.0204) \qquad\qquad \textbf{(21.10.1)} \\ R^2 &= 0.3360 \quad d = 2.0542 \end{aligned}$$

The 1 percent critical DF τ value is -3.4574. Since the computed $\tau\ (= t)$ of -11.0204 is more negative than the critical value, we conclude that the first-differenced LGDP is stationary; that is, it is $I(0)$. It is as shown in Figure 21.9. If you compare Figure 21.9 with Figure 21.1, you will see the obvious difference between the two.

FIGURE 21.9
First differences of logs of U.S. GDP, 1947–2007 (quarterly).



[38] A discussion of these tests can be found in Maddala et al., op. cit., Chapter 4.

[39] If a time series is $I(2)$, it will contain two unit roots, in which case we will have to difference it twice. If it is $I(d)$, it has to be differenced d times, where d is any integer.

Trend-Stationary Processes

As we have seen in Figure 21.5, a TSP is stationary around the trend line. Hence, the simplest way to make such a time series stationary is to regress it on time and the residuals from this regression will then be stationary. In other words, run the following regression:

$$Y_t = \beta_1 + \beta_2 t + u_t \tag{21.10.2}$$

where Y_t is the time series under study and where t is the trend variable measured chronologically.

Now

$$\hat{u}_t = (Y_t - \hat{\beta}_1 - \hat{\beta}_2 t) \tag{21.10.3}$$

will be stationary. $\hat{u}_t$ is known as a (linearly) **detrended time series.**

It is important to note that the trend may be nonlinear. For example, it could be

$$Y_t = \beta_1 + \beta_2 t + \beta_3 t^2 + u_t \tag{21.10.4}$$

which is a quadratic trend series. If that is the case, the residuals from Eq. (21.10.4) will now be (quadratically) detrended time series.

It should be pointed out that if a time series is DSP but we treat it as TSP, this is called **underdifferencing.** On the other hand, if a time series is TSP but we treat it as DSP, this is called **overdifferencing.** The consequences of these types of specification errors can be serious, depending on how one handles the serial correlation properties of the resulting error terms.[40]

To see what happens if we confuse a TSP series with a DSP series or vice versa, Figure 21.10 shows the first-differenced LGDP and the residuals of LGDP estimated from the TSP regression (21.10.2):

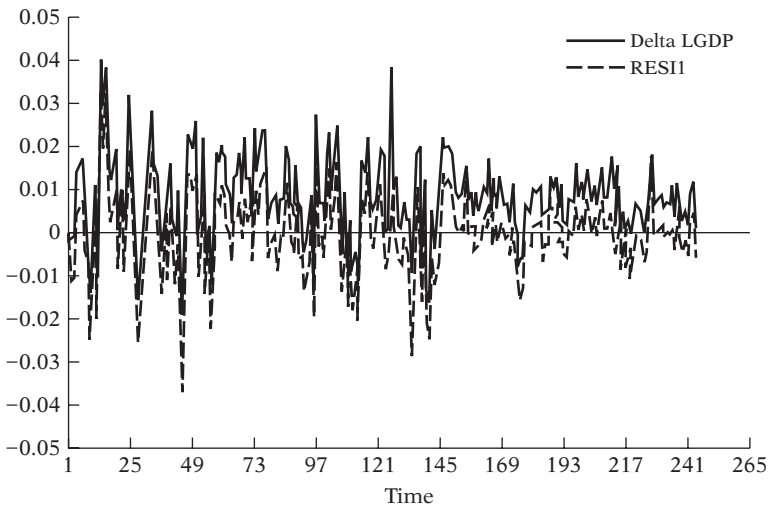


FIGURE 21.10
First differences (delta LGDP) and deviations from trend (RESI1) for logged GDP, 1947–2007 (quarterly).

[40]For a detailed discussion of this, see Maddala et al., op. cit., Section 2.7.

A look at this figure tells us that the first differences of real logged DGP are stationary (as confirmed by regression (21.10.1) but the residuals from the trend line (RESI1) are not.

In summary, ". . . it is very important to apply the right sort of stationarity transform to the data, if they are not already stationary. Most financial markets generate price, rate or yield data that are non-stationary because of stochastic rather than a deterministic trend. It is hardly ever appropriate to detrend the data by fitting a trend line and taking deviations. Instead the data should be detrended by taking first differences, usually of the log price or rates, because then the transformed stationary data will correspond to market returns."[41]

21.11 Cointegration: Regression of a Unit Root Time Series on Another Unit Root Time Series

We have warned that the regression of a nonstationary time series on another nonstationary time series may produce a spurious regression. Let us suppose that we consider the LPCE and LDPI time series data introduced in Section 21.1 (see the book's website for the actual data). Subjecting these time series individually to unit root analysis, you will find that they both are $I(1)$; that is, they contain a stochastic trend. It is quite possible that the two series *share the same common trend* so that the regression of one on the other will not be necessarily spurious.

To be specific, we use the U.S. economic time series data (see Section 21.1 and the book's website) and run the following regression of LPCE on LDPI:

$$\text{LPCE}_t = \beta_1 + \beta_2 \text{LDPI}_t + u_t \qquad \textbf{(21.11.1)}$$

where L denotes logarithm. β_2 is the elasticity of real personal consumption expenditure with respect to real disposable personal income. For illustrative purposes, we will call it *consumption elasticity.* Let us write this as:

$$u_t = \text{LPCE}_t - \beta_1 - \beta_2 \text{LDPI}_t \qquad \textbf{(21.11.2)}$$

Suppose we now subject u_t to unit root analysis and find that it is stationary; that is, it is $I(0)$. This is an interesting situation, for although LPCE_t and LDPI_t are individually $I(1)$, that is, they have stochastic trends, their linear combination (21.11.2) is $I(0)$. So to speak, the linear combination cancels out the stochastic trends in the two series. If you take consumption and income as two $I(1)$ variables, savings defined as (income − consumption) could be $I(0)$. As a result, a regression of consumption on income as in Eq. (21.11.1) would be meaningful (i.e., not spurious). In this case we say that the two variables are **cointegrated.** Economically speaking, two variables will be cointegrated if they have a long-term, or equilibrium, relationship between them. Economic theory is often expressed in equilibrium terms, such as Fisher's quantity theory of money or the theory of purchasing power parity (PPP), just to name a few.

In short, provided we check that the residuals from regressions like (21.11.1) are $I(0)$ or stationary, the traditional regression methodology (including the t and F tests) that we have considered extensively is applicable to data involving (nonstationary) time series. The valuable contribution of the concepts of unit root, cointegration, etc. is to force us to find out if the regression residuals are stationary. As Granger notes, "A test for cointegration can be thought of as a pre-test to avoid 'spurious regression' situations."[42]

In the language of cointegration theory, a regression such as Eq. (21.11.1) is known as a **cointegrating regression** and the slope parameter β_2 is known as the **cointegrating parameter.** The concept of cointegration can be extended to a regression model containing k regressors. In this case we will have k cointegrating parameters.

[41]Carol Alexander, op. cit., p. 324.

[42]C. W. J. Granger, "Developments in the Study of Co-Integrated Economic Variables," *Oxford Bulletin of Economics and Statistics,* vol. 48, 1986, p. 226.

Testing for Cointegration

A number of methods for testing cointegration have been proposed in the literature. We consider here a comparatively simple method, namely the DF or ADF unit root test on the residuals estimated from the cointegrating regression.[43]

Engle–Granger (EG) or Augmented Engle–Granger (AEG) Test

We already know how to apply the DF or ADF unit root tests. All we have to do is estimate a regression like Eq. (21.11.1), obtain the residuals, and use the DF or ADF tests.[44] There is one precaution to exercise, however. Since the estimated u_t are based on the *estimated* cointegrating parameter β_2, the DF and ADF critical significance values are not quite appropriate. Engle and Granger have calculated these values, which can be found in the references.[45] Therefore, the DF and ADF tests in the present context are known as **Engle–Granger (EG)** and **augmented Engle–Granger (AEG)** tests. However, several software packages now present these critical values along with other outputs.

Let us illustrate these tests. Using the data introduced in Section 21.1 and found on the book's website, we first regressed LPCEC on LDPIC and obtained the following regression:

$$\begin{aligned} \widehat{\text{LPCE}}_t &= \quad -0.1942 \; + \quad 1.0114\text{LDPI}_t \\ t &= \quad (-8.2328) \quad (348.5429) \qquad\qquad \textbf{(21.11.3)} \\ & \qquad\qquad R^2 = 0.9980 \qquad d = 0.1558 \end{aligned}$$

Since LPCE and LDPI are individually nonstationary, there is the possibility that this regression is spurious. But when we performed a unit root test on the residuals obtained from Eq. (21.11.3), we obtained the following results:

$$\begin{aligned} \widehat{\Delta \hat{u}_t} &= \; -0.0764\hat{u}_{t-1} \\ t &= (-3.0458) \qquad\qquad \textbf{(21.11.4)} \\ & \qquad R^2 = 0.0369 \qquad d = 2.5389 \end{aligned}$$

The Engle–Granger asymptotic 5 percent and 10 percent critical values are about −3.34 and −3.04, respectively. Therefore, the residuals from the regression are not stationary at the 5 percent level. It would be difficult to accept this reason, for economic theory suggests that there should be a stable relationship between PCE and DPI.

Let us reestimate Eq. (21.11.3) including the trend variable and then see if the residuals from this equation are stationary. We present the results first and then discuss what may be going on.

$$\begin{aligned} \widehat{\text{LPCE}}_t &= \quad 2.8130 + \quad 0.0037_t \; + \quad 0.5844\text{LDPI}_t \\ t &= (21.3491) \quad (22.9394) \quad (31.2754) \qquad\qquad \textbf{(21.11.3}\boldsymbol{a}\textbf{)} \\ & \qquad\qquad\qquad R^2 = 0.9994 \qquad d = 0.2956 \end{aligned}$$

[43]There is this difference between tests for unit roots and tests for cointegration. As David A. Dickey, Dennis W. Jansen, and Daniel I. Thornton observe, "Tests for unit roots are performed on univariate [i.e., single] time series. In contrast, cointegration deals with the relationship among a group of variables, where (unconditionally) each has a unit root." See their article, "A Primer on Cointegration with an Application to Money and Income," *Economic Review,* Federal Reserve Bank of St. Louis, March–April 1991, p. 59. As the name suggests, this article is an excellent introduction to cointegration testing.

[44]If PCE and DPI are not cointegrated, any linear combination of them will be nonstationary and, therefore, the u_t will also be nonstationary.

[45]R. F. Engle and C. W. Granger, "Co-integration and Error Correction: Representation, Estimation and Testing," *Econometrica,* vol. 55, 1987, pp. 251–276.

To see if the residuals from this regression are stationary, we obtained the following results (compare with Eq. [21.11.4]):

$$\widehat{\Delta u_t} = -0.1498\hat{u}_{t-1}$$
$$t = (-4.4545) \qquad \textbf{(21.11.4}a\textbf{)}$$
$$R^2 = 0.0758 \qquad d = 2.3931$$

Note: $\hat{u}_t$ is the residual from Eq. (21.11.3*a*).

The DF test now shows that these residuals are stationary. Even if we use ADF with several lags, the residuals are still stationary.

What is going on here? Although the residuals from regression (21.11.4*a*) are stationary, that is, they are $I(0)$, they are stationary around a deterministic time trend, the trend here being linear. That is, the residuals are $I(0)$ plus a linear trend. As noted earlier, a time series may contain both a deterministic and a stochastic trend.

Before we proceed further, it should be noted that our time series data cover a long period of time (61 years). It is quite possible that because of structural changes in the U.S. economy over this period, our results and conclusions are likely to differ. In Exercise 21.28 you are asked to check for this possibility.

Cointegration and Error Correction Mechanism (ECM)

We just showed that, allowing for the (linear) trend, LPCE and LDPI seem to be cointegrated, that is, there is a long-term, or equilibrium, relationship between the two. Of course, in the short-run there may be disequilibrium. Therefore, we can treat the error term in the following equation as the "equilibrium error." And we can use this error term to tie the short-run behavior of PCE to its long-run value:

$$u_t = \text{LPCE}_t - \beta_1 - \beta_2\text{LDPI} - \beta_3 t \qquad \textbf{(21.11.5)}$$

The **error correction mechanism (ECM)** first used by Sargan[46] and later popularized by Engle and Granger corrects for disequilibrium. An important theorem, known as the **Granger representation theorem,** states that if two variables Y and X are cointegrated, the relationship between the two can be expressed as ECM. To see what this means, let us revert to our PCE–DPI example. Now consider the following model:

$$\Delta\text{LPCE}_t = \alpha_0 + \alpha_1\Delta\text{LDPI}_t + \alpha_2\text{u}_{t-1} + \varepsilon_t \qquad \textbf{(21.11.6)}$$

where ε_t is a white noise error term and u_{t-1} is the lagged value of the error term in Eq. (21.11.5).

ECM equation (21.11.5) states that ΔLPCE depends on ΔLDPI and also on the equilibrium error term.[47] If the latter is nonzero, then the model is out of equilibrium. Suppose ΔLDPI is zero and u_{t-1} is positive. This means LPCE_{t-1} is too high to be in equilibrium, that is, LPCE_{t-1} is above its equilibrium value of $(\alpha_0 + \alpha_1\text{LDPI}_{t-1})$. Since α_2 is expected to be negative, the term $\alpha_2 u_{t-1}$ is negative and, therefore, ΔLPCE_t will be negative to restore the equilibrium. That is, if LPCE_t is above its equilibrium value, it will start falling in the next period to correct the equilibrium error; hence the name ECM. By the same token, if u_{t-1} is negative (i.e., LPCE is below its equilibrium value), $\alpha_2 u_{t-1}$ will be positive, which will cause ΔLPCE_t to be positive, leading LPCE_t to rise in period t. Thus, the absolute value of α_2 decides how quickly the equilibrium is restored. In practice, we estimate u_{t-1} by

[46]J. D. Sargan, "Wages and Prices in the United Kingdom: A Study in Econometric Methodology," in K. F. Wallis and D. F. Hendry, eds., *Quantitative Economics and Econometric Analysis,* Basil Blackwell, Oxford, U.K., 1984.

[47]The following discussion is based on Gary Koop, op. cit., pp. 159–160 and Kerry Peterson, op. cit., Section 8.5.

$\hat{u}_{t-1} = (\text{LPCE}_t - \hat{\beta}_1 - \hat{\beta}_2\text{LDPI} - \hat{\beta}_3 t)$. Keep in mind that the error correction coefficient α_2 is expected to be negative (why?).

Returning to our illustrative example, the empirical counterpart of Eq. (21.11.6) is:

$$\begin{aligned} \widehat{\Delta\text{LPCE}}_t = \ & 0.0061 + 0.2967\Delta\text{LDPI}_t - 0.1223\hat{u}_{t-1} \\ t = \ & (9.6753) \quad (6.2282) \quad (-3.8461) \\ & R^2 = 0.1658 \quad d = 2.1496 \end{aligned} \tag{21.11.7}$$

Statistically, the ECM term is significant, suggesting that PCE adjusts to DPI with a lag; only about 12 percent of the discrepancy between long-term and short-term PCE is corrected within a quarter.

From regression (21.11.7) we see that the short-run consumption elasticity is about 0.29. The long-run elasticity is about 0.58, which can be seen from Eq. (21.11.3*a*).

Before we conclude this section, the caution sounded by S. G. Hall is worth remembering:

> While the concept of cointegration is clearly an important theoretical underpinning of the error correction model there are still a number of problems surrounding its practical application; the critical values and small sample performance of many of these tests are unknown for a wide range of models; informed inspection of the correlogram may still be an important tool.[48]

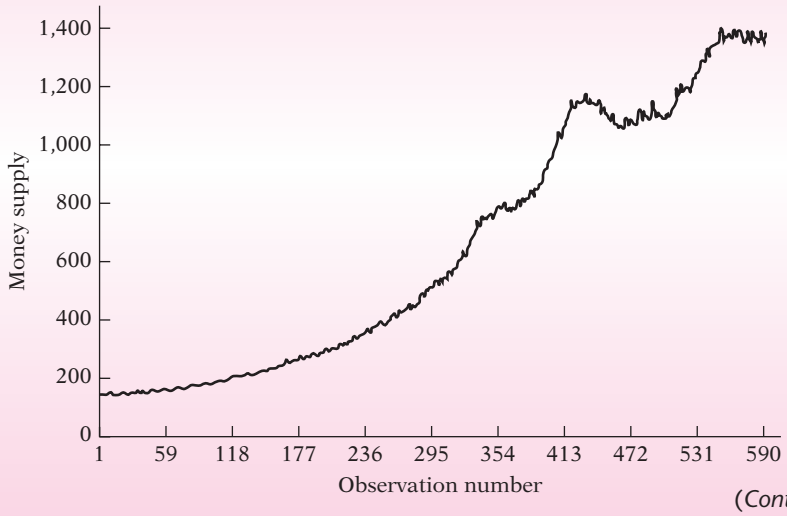
21.12 Some Economic Applications

We conclude this chapter by considering some concrete examples.

EXAMPLE 21.1
M1 Monthly Money Supply in the United States, January 1959 to March 1, 2008

Figure 21.11 shows the M1 money supply for the United States from January 1959 to March 1, 2008. From our knowledge of stationarity, it seems that the M1 money supply time series is nonstationary, which can be confirmed by unit root analysis. (*Note:* To save

FIGURE 21.11 U.S. money supply over 1959:01 to 2008:03.



(*Continued*)

[48]S. G. Hall, "An Application of the Granger and Engle Two-Step Estimation Procedure to the United Kingdom Aggregate Wage Data," *Oxford Bulletin of Economics and Statistics,* vol. 48, no. 3, August 1986, p. 238. See also John Y. Campbell and Pierre Perron, "Pitfalls and Opportunities: What Macroeconomists Should Know about Unit Roots," NBER (National Bureau of Economic Research) *Macroeconomics Annual 1991,* pp. 141–219.

EXAMPLE 21.1 (*Continued*)

space, we have not given the actual data, which can be obtained from the Federal Reserve Board or the Federal Reserve Bank of St. Louis.)

$$\begin{aligned} \Delta\hat{M}_t = \ & -0.1347 + 0.0293t - 0.0102M_{t-1} \\ t = \ & (-0.14) \quad (2.62) \quad (-2.30) \\ & R^2 = 0.0130 \quad d = 2.2325 \end{aligned} \tag{21.12.1}$$

The 1, 5, and 10 percent critical τ values are -3.9811, -3.4210, and -3.1329. Since the t value of -2.30 is less negative than any of these critical values, the conclusion is that the M1 time series is nonstationary; that is, it contains a unit root or it is $I(1)$. Even when several lagged values of ΔM_t (à la ADF) were introduced, the conclusion did not change. On the other hand, the first differences of the M1 money supply were found to be stationary (check this out).

EXAMPLE 21.2
The U.S./U.K. Exchange Rate: January 1971 to April 2008

Figure 21.12 gives the graph of the (\$/£) exchange rate from January 1971 to April 2008, for a total of 286 observations. By now you should be able to spot this time series as nonstationary. Carrying out the unit root tests, we obtained the following τ statistics: -0.82 (no intercept, no trend), -1.96 (intercept), and -1.33 (intercept and trend). Each of these statistics, in absolute value, was less than its critical τ value from the appropriate DF tables, thus confirming the graphical impression that the U.S./U.K. exchange rate time series is nonstationary.

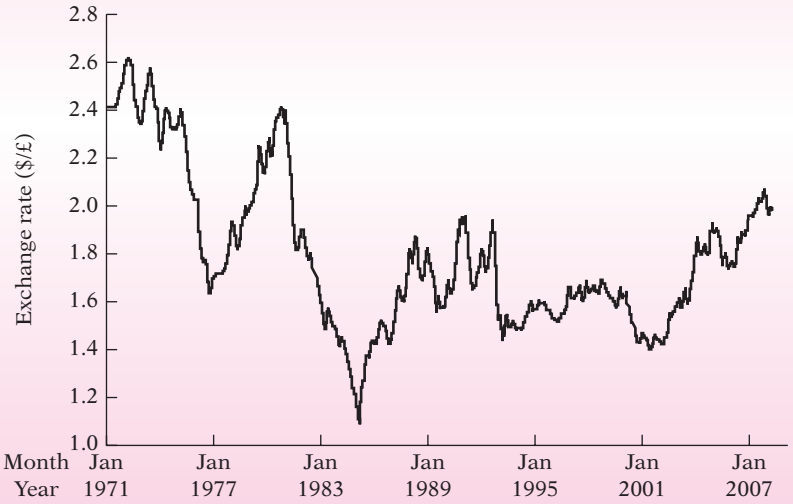


FIGURE 21.12
U.S./U.K. exchange rate: January 1971 to April 2008.

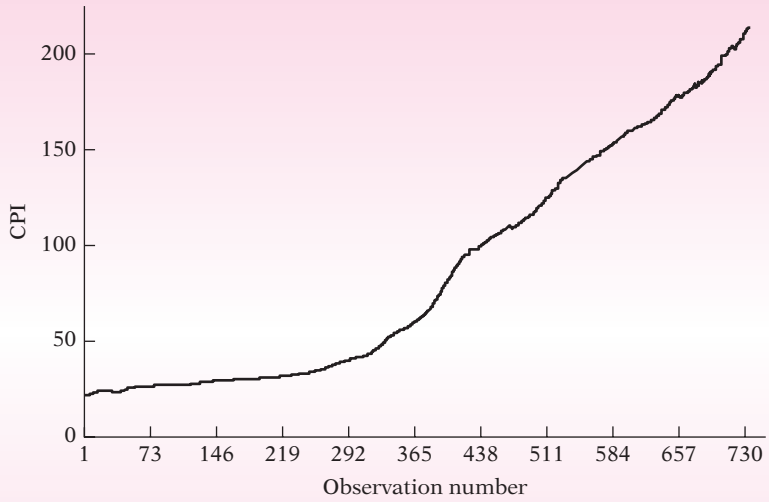
EXAMPLE 21.3
U.S. Consumer Price Index (CPI), January 1947 to March 2008

Figure 21.13 shows the U.S. CPI from January 1947 to March 2008 for a total of 733 observations. The CPI series, like the M1 series considered previously, shows a sustained upward trend. The unit root exercise gave the following results:

$$\begin{aligned} \widehat{\Delta \text{CPI}}_t = \ & -0.01082 + 0.00068t - 0.00096\text{CPI}_{t-1} + 0.40669\Delta\text{CPI}_{t-1} \\ t = \ & (-0.54) \quad (4.27) \quad (-1.77) \quad (12.03) \\ & R^2 = 0.3570 \quad d = 1.9295 \end{aligned} \tag{21.12.2}$$

EXAMPLE 21.3
(*Continued*)

FIGURE 21.13 U.S. CPI, January 1947 to March 2008.

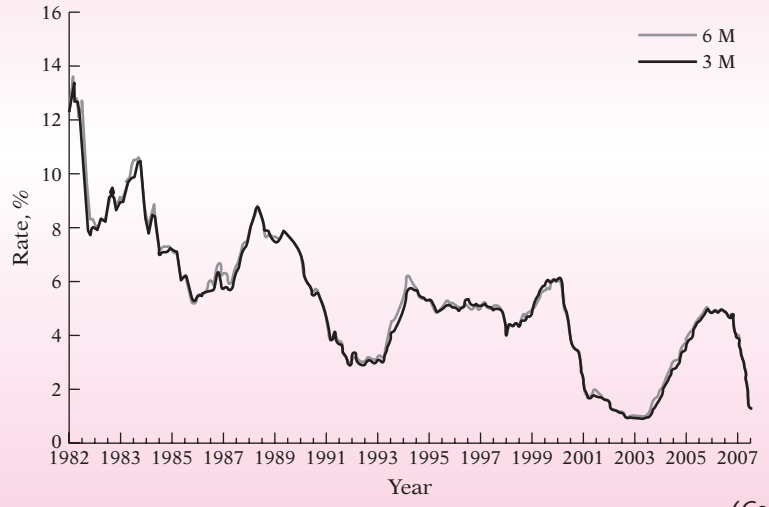


The t ($= \tau$) value of CPI_{t-1} is -1.77. The 10 percent critical value is -3.1317. Since, in absolute terms, the computed τ is less than the critical τ, the conclusion is that CPI is not a stationary time series. We can characterize it as having a stochastic trend (why?). However, if you take the first differences of the CPI series, you will find them to be stationary. Hence CPI is a difference-stationary (DS) time series.

EXAMPLE 21.4
Are 3-Month and 6-Month Treasury Bill Rates Cointegrated?

Figure 21.14 plots (constant maturity) 3-month and 6-month U.S. Treasury bill (T-bill) rates from January 1982 to March 2008, for a total of 315 observations. Does the graph show that the two rates are cointegrated; that is, is there an equilibrium relationship between the two? From financial theory, we would expect that to be the case, otherwise arbitrageurs will exploit any discrepancy between the short and the long rates. First of all, let us see if the two time series are stationary.

FIGURE 21.14
Three- and six-month Treasury bill rates (constant maturity).



(*Continued*)

EXAMPLE 21.4
(*Continued*)

On the basis of the pure random walk model (i.e., no intercept, no trend), both the rates were stationary. Including intercept, trend, and one lagged difference, the results suggested that the two rates might be trend stationary; the trend coefficient in both cases was negative and significant at about the 7 percent level. So, depending on which results we accept, the two rates are either stationary or trend stationary.

Regressing the 6-month T-bill rate (TB6) on the 3-month T-bill rate, we obtained the following regression.

$$\begin{aligned} \widehat{\text{TB6}}_t &= 0.0842 + 1.0078\text{TB3}_t \\ t &= (3.65) \quad (252.39) \\ & \qquad R^2 = 0.995 \qquad d = 0.4035 \end{aligned} \tag{21.12.3}$$

Applying the unit root test to the residuals from the preceding regression, we found that the residuals were stationary, suggesting that the 3- and 6-month T-bill rates were cointegrated. Using this knowledge, we obtained the following error correction model (ECM):

$$\begin{aligned} \widehat{\Delta\text{TB6}}_t &= -0.0047 + 0.8992\Delta\text{TB3}_t - 0.1855\hat{u}_{t-1} \\ t &= (-0.82) \quad (47.77) \quad (-5.69) \\ & \qquad R^2 = 0.880 \qquad d = 1.5376 \end{aligned} \tag{21.12.4}$$

where $\hat{u}_{t-1}$ is the lagged value of the error correction term from the preceding period. As these results show, 0.19 of the discrepancy in the two rates in the previous month is eliminated this month.[49] Besides, short-run changes in the 3-month T-bill rate are quickly reflected in the 6-month T-bill rate, as the slope coefficient between the two is 0.8992. This should not be a surprising finding in view of the efficiency of the U.S. money markets.

Summary and Conclusions

1. Regression analysis based on time series data implicitly assumes that the underlying time series are stationary. The classical *t* tests, *F* tests, etc., are based on this assumption.
2. In practice most economic time series are nonstationary.
3. A stochastic process is said to be **weakly stationary** if its mean, variance, and autocovariances are constant over time (i.e., they are time-invariant).
4. At the informal level, weak stationarity can be tested by the **correlogram** of a time series, which is a graph of autocorrelation at various lags. For stationary time series, the correlogram tapers off quickly, whereas for nonstationary time series it dies off gradually. For a purely random series, the autocorrelations at all lags 1 and greater are zero.
5. At the formal level, stationarity can be checked by finding out if the time series contains a unit root. The **Dickey–Fuller (DF)** and **augmented Dickey–Fuller (ADF)** tests can be used for this purpose.
6. An economic time series can be **trend stationary (TS)** or **difference stationary (DS).** A TS time series has a deterministic trend, whereas a DS time series has a variable, or stochastic, trend. The common practice of including the time or trend variable in a regression model to detrend the data is justifiable only for TS time series. The DF and ADF tests can be applied to determine whether a time series is TS or DS.

[49]Since both T-bill rates are in percent form, this would suggest that if the 6-month TB rate was higher than the 3-month TB rate more than expected a priori in the last month, this month it will be reduced by 0.19 percentage points to restore the long-run relationship between the two interest rates. For the underlying theory about the relationship between short- and long-run interest rates, see any money and banking textbook and read up on the term structure of interest rates.

7. Regression of one time series variable on one or more time series variables often can give nonsensical or spurious results. This phenomenon is known as **spurious regression.** One way to guard against it is to find out if the time series are cointegrated.
8. **Cointegration** means that despite being individually nonstationary, a linear combination of two or more time series can be stationary. The **Engle–Granger (EG)** and the **augmented Engle–Granger (AEG)** tests can be used to find out if two or more time series are cointegrated.
9. Cointegration of two (or more) time series suggests that there is a long-run, or equilibrium, relationship between them.
10. The **error correction mechanism (ECM)** developed by Engle and Granger is a means of reconciling the short-run behavior of an economic variable with its long-run behavior.
11. The field of time series econometrics is evolving. The established results and tests are in some cases tentative and a lot more work remains. An important question that needs an answer is why some economic time series are stationary and others are nonstationary.

EXERCISES

Questions

21.1. What is meant by weak stationarity?
21.2. What is meant by an integrated time series?
21.3. What is the meaning of a unit root?
21.4. If a time series is $I(3)$, how many times would you have to difference it to make it stationary?
21.5. What are Dickey–Fuller (DF) and augmented DF tests?
21.6. What are Engle–Granger (EG) and augmented EG tests?
21.7. What is the meaning of cointegration?
21.8. What is the difference, if any, between tests of unit roots and tests of cointegration?
21.9. What is spurious regression?
21.10. What is the connection between cointegration and spurious regression?
21.11. What is the difference between a deterministic trend and a stochastic trend?
21.12. What is meant by a trend-stationary process (TSP) and a difference-stationary process (DSP)?
21.13. What is a random walk (model)?
21.14. "For a random walk stochastic process, the variance is infinite." Do you agree? Why?
21.15. What is the error correction mechanism (ECM)? What is its relationship with cointegration?

Empirical Exercises

21.16. Using the U.S. economic time series data posted on the book's website, obtain sample correlograms up to 36 lags for the time series LPCE, LDPI, LCP(profits), and LDIVIDENDS. What general pattern do you see? Intuitively, which one(s) of these time series seems to be stationary?
21.17. For each of the time series of Exercise 21.16, use the DF test to find out if these series contain a unit root. If a unit root exists, how would you characterize such a time series?

21.18. Continue with Exercise 21.17. How would you decide if the ADF test is more appropriate than the DF test?

21.19. Consider the dividends and profits time series given in the U.S. economic time series data posted on the book's website. Since dividends depend on profits, consider the following simple model:

$$\text{LDIVIDENDS}_t = \beta_1 + \beta_2 \text{LCP} + u_t$$

a. Would you expect this regression to suffer from the spurious regression phenomenon? Why?

b. Are the logged Dividends and logged Profits time series cointegrated? How do you test for this explicitly? If, after testing, you find that they are cointegrated, would your answer in (*a*) change?

c. Employ the error correction mechanism (ECM) to study the short- and long-run behavior of dividends in relation to profits.

d. If you examine the LDIVIDENDS and LCP series individually, do they exhibit stochastic or deterministic trends? What tests do you use?

e. Assume LDIVIDENDS and LCP are cointegrated. Then, instead of regressing dividends on profits, you regress profits on dividends. Is such a regression valid?

21.20. Take the first differences of the time series given in the U.S. economic time series data posted on the book's website and plot them. Also obtain a correlogram of each time series up to 36 lags. What strikes you about these correlograms?

21.21. Instead of regressing LDIVIDENDS on LCP in level form, suppose you regress the first difference of LDIVIDENDS on the first difference of LCP. Would you include the intercept in this regression? Why or why not? Show the calculations.

21.22. Continue with the previous exercise. How would you test the first-difference regression for stationarity? In the present example, what would you expect a priori and why? Show all the calculations.

21.23. From the U.K. private sector housing starts (X) for the period 1948 to 1984, Terence Mills obtained the following regression results:†

$$\begin{aligned} \widehat{\Delta X}_t &= \quad 31.03 \; - \quad 0.188 X_{t-1} \\ \text{se} &= (12.50) \qquad (0.080) \\ (t =)\tau &\qquad\quad (-2.35) \end{aligned}$$

Note: The 5 percent critical τ value is -2.95 and the 10 percent critical τ value is -2.60.

a. On the basis of these results, is the housing starts time series stationary or nonstationary? Alternatively, is there a unit root in this time series? How do you know?

b. If you were to use the usual t test, is the observed t value statistically significant? On this basis, would you have concluded that this time series is stationary?

c. Now consider the following regression results:

$$\begin{aligned} \widehat{\Delta^2 X_t} &= \; 4.76 \; - 1.39 \Delta X_{t-1} + 0.313 \Delta^2 X_{t-1} \\ \text{se} &= (5.06) \quad (0.236) \qquad (0.163) \\ (t =)\tau &\qquad (-5.89) \end{aligned}$$

*Optional.

†Terence C. Mills, op. cit., p. 127. Notation slightly altered.

where Δ^2 is the second difference operator, that is, the first difference of the first difference. The estimated τ value is now statistically significant. What can you say now about the stationarity of the time series in question?

Note: The purpose of the preceding regression is to find out if there is a **second unit root** in the time series.

21.24. Generate two random walk series as indicated in Eqs. (21.7.1) and (21.7.2) and regress one on the other. Repeat this exercise but now use their first differences and verify that in this regression the R^2 value is about zero and the Durbin–Watson d is close to 2.

21.25. To show that two variables, each with deterministic trend, can lead to spurious regression, Charemza et al. obtained the following regression based on 30 observations:*

$$\hat{Y}_t = 5.92 + 0.030X_t$$
$$t = (9.9) \quad (21.2)$$
$$R^2 = 0.92 \qquad d = 0.06$$

where $Y_1 = 1, Y_2 = 2, \ldots, Y_n = n$ and $X_1 = 1, X_2 = 4, \ldots, X_n = n^2$.

a. What kind of trend does Y exhibit? and X?

b. Plot the two variables and plot the regression line. What general conclusion do you draw from this plot?

21.26. From the data for the period 1971–I to 1988–IV for Canada, the following regression results were obtained:

1. $$\widehat{\ln \text{M1}}_t = -10.2571 + 1.5975 \ln \text{GDP}_t$$
$$t = (-12.9422) \quad (25.8865)$$
$$R^2 = 0.9463 \qquad d = 0.3254$$

2. $$\widehat{\Delta \ln \text{M1}}_t = 0.0095 + 0.5833 \Delta \ln \text{GDP}_t$$
$$t = (2.4957) \quad (1.8958)$$
$$R^2 = 0.0885 \qquad d = 1.7399$$

3. $$\Delta \hat{u}_t = -0.1958 \hat{u}_{t-1}$$
$$(t = \tau) \; (-2.2521)$$
$$R^2 = 0.1118 \qquad d = 1.4767$$

where M1 = M1 money supply, GDP = gross domestic product, both measured in billions of Canadian dollars, ln is natural log, and $\hat{u}_t$ represent the estimated residuals from regression (1).

a. Interpret regressions (1) and (2).

b. Do you suspect that regression (1) is spurious? Why?

c. Is regression (2) spurious? How do you know?

d. From the results of regression (3), would you change your conclusion in (*b*)? Why?

*Charemza et al., op. cit., p. 93.

e. Now consider the following regression:

$$\widehat{\Delta \ln M1_t} = 0.0084 + 0.7340 \Delta \ln GDP_t - 0.0811 \hat{u}_{t-1}$$
$$t = (2.0496) \quad (2.0636) \quad (-0.8537)$$
$$R^2 = 0.1066 \qquad d = 1.6697$$

What does this regression tell you? Does this help you decide if regression (1) is spurious or not?

21.27. The following regressions are based on the CPI data for the United States for the period 1960–2007, for a total of 48 annual observations:

1. $$\widehat{\Delta CPI_t} = 0.0334 CPI_{t-1}$$
$$t = (12.37)$$
$$R^2 = 0.0703 \qquad d = 0.3663 \qquad RSS = 206.65$$

2. $$\widehat{\Delta CPI_t} = 1.8662 + 0.0192 CPI_{t-1}$$
$$t = (3.27) \quad (3.86)$$
$$R^2 = 0.249 \qquad d = 0.4462 \qquad RSS = 166.921$$

3. $$\widehat{\Delta CPI_t} = 1.1611 + 0.5344t - 0.1077 CPI_{t-1}$$
$$t = (2.37) \quad (4.80) \quad (-4.02)$$
$$R^2 = 0.507 \qquad d = 0.6071 \qquad RSS = 109.608$$

where RSS = residual sum of squares.

a. Examining the preceding regressions, what can you say about stationarity of the CPI time series?

b. How would you choose among the three models?

c. Equation (1) is Eq. (3) minus the intercept and trend. Which test would you use to decide if the implied restrictions of model (1) are valid? (*Hint:* Use the Dickey–Fuller *t* and *F* tests. Use the approximate values given in **Appendix D,** Table D.7.)

21.28. As noted in the text, there may be several structural breaks in the U.S. economic time series dataset introduced in Section 21.1. Dummy variables are a good way of incorporating these shifts in the data.

a. Using dummy variables to designate three different periods related to the oil embargoes in 1973 and 1979, regress the log of personal consumption expenditures (LPCE) on the log of disposable personal income (LDPI). Has there been a change in the results? What is your decision about the unit root hypothesis now?

b. Several websites list the official economic cycles that may have affected the U.S. economic time series data discussed in Section 21.1. See, for example, http://www.nber.org/cycles/cyclesmain.html. Using the information here, create dummy variables indicating some of the major cycles and check the results of regressing LPCE on LDPI. Has there been a change?